AN ASYMPTOTIC COUNT OF EULER PRODUCTS

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Abstract. For every fixed positive integer N with $\mathrm{rad}(N) > \sqrt{N}$, we prove that the set $\mathcal{E}(N,k)$ of normalized degree-two Euler products with the prescribed ramified factors and an entire untwisted weight-k functional equation is finite, and that

$$\#\mathcal{E}(N,k) = \dim M_k^\circ(\Gamma_0(N)) + O_N\big(k/\log\log(2k)\big)$$

for positive even k. Here M_k° is the two-trace new subspace of modular forms vanishing at 0 and ∞. The argument combines strict reconstruction from complete ramified rows, a regular trace calculation, detecting prime blocks and a quantitative holomorphic trace comparison. The reconstruction, local calculation, prime preparation and endpoint schedule are developed together below. The augmented spanning theorem is the combinatorial input from the accompanying paper. No fixed coefficient-growth exponent is assumed.

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1. Statement and normalization

Let $k \geq 2$ be even. The set $\mathcal{E}(N,k)$ consists of sequences $f = (f_n)_{n\geq 1}$ with a Dirichlet series converging in a right half-plane,

$$L_f(s) = \sum_{n\geq 1} f_n n^{-s} = \prod_p (1 - f_p p^{-s} + \mathbf{1}_{p\nmid N} p^{-2s})^{-1}, \qquad f_p^2 = \begin{cases} p^{-1}, & p \parallel N, \\ 0, & p^2 \mid N, \end{cases}$$

such that

$$\Lambda_f(s) = N^{(s-1/2)/2}\, 2(2\pi)^{-s-(k-1)/2}\Gamma(s + (k-1)/2) L_f(s)$$

is entire of finite order and $\Lambda_f(s) = \epsilon_f \Lambda_f(1-s)$, with $\epsilon_f \in \{\pm 1\}$. All coefficients f_n throughout are normalized; the critical line is always $\Re s = 1/2$. Write

$$F(z) = \sum_{n\geq 1} f_n n^{(k-1)/2} e^{2\pi i n z}.$$

Convergence somewhere implies a polynomial coefficient bound, with an exponent initially allowed to depend on f.

Use determinant-normalized weight-k slash operators and $W_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$. Define

$$M_k^{\mathrm{new}}(N) = \{F \in M_k(\Gamma_0(N)) : \mathrm{Tr}^N_{N/p} F = \mathrm{Tr}^N_{N/p}(F|_k W_N) = 0 \ (p \mid N)\},$$

and let $M_k^\circ(N)$ be its subspace vanishing at 0 and ∞.

Theorem 1.2 (Theorem 1.2 of the accompanying paper)**.** *If* $\mathrm{rad}(N) > \sqrt{N}$*, then* $\mathcal{E}(N,k)$ *is finite for every positive even* k*, and*

$$\#\mathcal{E}(N,k) = \dim M_k^\circ(N) + O_N\left(\frac{k}{\log\log(2k)}\right).$$

The implied constant is independent of k *and of all solution-dependent growth bounds. The same assertion holds with* $\dim S_k^{\mathrm{new}}(N)$ *in place of* $\dim M_k^\circ(N)$*.*

The big-O assertion is along the positive even integers. Finiteness is proved at every such weight, separately from the asymptotic assertion.

All logarithms are natural. We shall use two representations of the same rational-branch expressions. The physical representation acts on weighted sections $y^{k/2}F(z)$ on the translation cylinder. The regular representation acts on square-summable functions on rational orbits and has a rooted trace. Holomorphic evaluation vectors are never asserted to be square summable on those orbits.

The determinant-normalized action is

$$(F|_k g)(z) = \det(g)^{k/2}(cz+d)^{-k} F(gz), \qquad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{R}).$$

Put $T_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, $W_L = \begin{pmatrix} 0 & -1 \\ L & 0 \end{pmatrix}$, and $\ell_N(q) = L_q = \mathrm{lcm}(N, q^2)$. Positive scalar matrices act trivially. Constants may depend on N and on the fixed finite choices in the proof, but are chosen before the weight and the iteration horizon.

2. The linear equations and strict reconstruction

Put $L_q = \mathrm{lcm}(N, q^2)$ and define

$$R_q F(z) = \sum_{\substack{a \bmod q \\ (a,q)=1}} F(z + a/q), \qquad C_q F(z) = \sum_{a \bmod q} F(z + a/q).$$

The Euler-factor calculation and Mellin inversion below give

$$R_qF(z) = w_q(\sqrt{L_q}z)^{-k}R_qF(-1/(L_qz)), \qquad w_q = w\prod_{\substack{p\mid q\\ p\| N}}(-\delta_p), \tag{2.1}$$

where $w = i^{-k}\epsilon_f$ and $\delta_p = \sqrt{p}\, f_p \in \{\pm 1\}$. These derivations hold at every level. If $p^2 \mid N$ and $p^2 \mid q$, the corresponding Ramanujan twist is identically zero, which causes no exception to the equation.

Here are the calculation and analytic justification. If $p^a \| q$, write $f_{p^j} = 0$ for $j < 0$, and set

$$D_{p,a}(s) = (1 - f_p p^{-s} + \mathbf{1}_{p\nmid N}p^{-2s})\sum_{j\ge 0} f_{p^j}c_{p^a}(p^j)p^{-js}.$$

For $p \nmid N$, direct summation gives

$$D_{p,a}(s) = p^{a(1-s)}\{f_{p^a} + p^{-1}f_{p^{a-2}} - f_{p^{a-1}}(p^{s-1} + p^{-s})\}, \quad D_{p,a}(s) = p^{a(1-2s)}D_{p,a}(1-s).$$

For $p \mid N$ it gives

$$D_{p,a}(s) = p^{(a-1)(1-s)}f_p^{a-1}(f_p p^{1-s} - 1).$$

When $p \| N$, the reflection multiplier is $-\delta_p p^{(2a-1)(1/2-s)}$. When $p^2 \mid N$, the factor is -1 for $a = 1$ and zero for $a \ge 2$. Multiplicativity now yields $L_f(s, c_q) = L_f(s)\prod_{p^a\| q} D_{p,a}(s)$. Its completion with conductor L_q is entire of finite order, with root number $\epsilon_f \prod_{p\mid q, p\| N}(-\delta_p)$. On a sufficiently far right vertical line, Mellin inversion gives

$$\frac{1}{2\pi i}\int_{(c)} \Lambda_f(s, c_q)(\sqrt{L_q}y)^{1/2-s}\, ds = 2y^{k/2}R_qF(iy).$$

Absolute convergence and Stirling's bound give exponential decay on that line; the functional equation gives the same bound on the reflected line. The finite-order Phragmén–Lindelöf argument between them permits the contour shift without residues. Substituting $s \mapsto 1 - s$ therefore gives the stated relation on the imaginary axis. Holomorphy extends it to $\mathbb{H}$, proving (2.1).

For a fixed sign vector $e = (w, (\delta_p)_{p\| N})$, let $V_k(e)$ be the space of periodic holomorphic functions on $\mathbb{H}$, vanishing uniformly at infinity, satisfying (2.1) and the normalized ramified recurrences

$$f_{pn} = \delta_p p^{-1/2}f_n \quad (p \| N), \qquad f_{pn} = 0 \quad (p^2 \mid N). \tag{2.2}$$

Polynomial growth is not imposed in this definition. Periodicity and vanishing give a positive Fourier expansion in which (2.2) is meaningful. There are $2^{1+\#\{p:p\| N\}}$ sectors.

Lemma 2.1 (The complete ramified rows)**.** *For every $F \in V_k(e)$,*

$$C_pF(z) = \begin{cases} \delta_p p^{k/2}F(pz), & p \| N, \\ 0, & p^2 \mid N. \end{cases} \tag{2.3}$$

In determinant-normalized slash notation the first right-hand side is $\delta_p F|_k \operatorname{diag}(p, 1)$. Its cost after weighting by $y^{k/2}$ is one.

Proof. Summing the Fourier expansion over translations kills all indices not divisible by p. At index pn, (2.2) gives $pf_{pn}(pn)^{(k-1)/2} = \delta_p p^{k/2}f_n n^{(k-1)/2}$ in the first case and zero in the second. Also $y^{k/2}p^{k/2}|F(pz)| = (py)^{k/2}|F(pz)|$. □

We use the following augmented spanning theorem from the accompanying paper [1, Section 4]: if $\mathcal{P}$ is a finite set of primes with product greater than $\sqrt{N}$, there is a positive integer Q, divisible by all of them, for which the legal Ramanujan rows together with all complete p-rows, $p \in \mathcal{P}$, span $\mathbb{Q}^{\mathbb{Z}/Q\mathbb{Z}}$. The legal window and the resulting certificate are displayed next.

Take $\mathcal{P} = \{p : p \mid N\}$. Its product exceeds $\sqrt{N}$, so augmented spanning supplies Q and rational certificates

$$e_j = \sum_{q,u} A_j(q,u) R^Q_{q,u} + \sum_{p\mid N,u} B_j(p,u) C^Q_{p,u} \qquad (0 \le j < Q), \tag{2.4}$$

with finite supports, where

$$R^Q_{q,u} = \sum_{(a,q)=1} e_{u+aQ/q}, \qquad C^Q_{p,u} = \sum_{a \bmod p} e_{u+aQ/p},$$

and every ordinary row used satisfies

$$-Q/\sqrt{L_q} < u < Q/\sqrt{L_q} - 1. \tag{2.5}$$

All data are fixed once N is fixed. Choose $\theta < 1$ so that $\sqrt{L_q}\max(|u|, |u+1|)/Q \le \theta$ on these finite supports. If ordinary rows occur, put

$$y_0 = \sqrt{\frac{1-\theta^2}{2\max L_q}}, \qquad \rho = \min\left(\frac{2}{1+\theta^2}, \min_{p\|N} p\right) > 1,$$

where an empty prime minimum is omitted. If no ordinary row occurs, take $y_0 = 1$ and any $1 < \rho \le 2$. Finally put

$$C = \max\left(1, \max_j \left[\sum_{q,u} |A_j(q,u)|\varphi(q) + \sum_{p\|N,u} |B_j(p,u)|\right]\right), \quad \gamma = \frac{\log C}{\log \rho}, \quad \nu = k/2 + \gamma. \tag{2.6}$$

Proposition 2.2 (Uniform reconstruction and cutoff)**.** *For every sector and $F \in V_k(e)$,*

$$\sup_{x\in\mathbb{R}, y>0} y^\nu |F(x+iy)| = \sup_{x\in\mathbb{R}, y\ge y_0} y^\nu |F(x+iy)| < \infty. \tag{2.7}$$

The first

$$B_k = \left\lfloor \frac{\nu \log 2}{\pi y_0} \right\rfloor = O_N(k) \tag{2.8}$$

Fourier coefficients determine F. In particular every $V_k(e)$ is finite dimensional, and every $F \in V_k(e)$ automatically satisfies $f_n = O_F(n^{\gamma+1/2})$.

Proof. Reduce x modulo one and choose j with $j/Q \le x \le (j+1)/Q$. Evaluate (2.4) on $F(z + (n-j)/Q)$. For the row (q,u), put $\beta = z + (u-j)/Q$ and use (2.1). Its new arguments have height

$$y' = \frac{y}{L_q|\beta|^2} \ge \frac{2y}{1+\theta^2} \ge \rho y \qquad (0 < y \le y_0).$$

The multiplier has absolute value $(y'/y)^{k/2}$, so its weighted cost is $\varphi(q)$. Lemma 2.1 handles the complete rows: they give a height-py branch of cost one or no branch at all. Thus the sum of weighted branch costs is at most C. Multiplying by y^γ makes this at most $C\rho^{-\gamma} = 1$ relative to the maximum of the new $(y')^\nu |F(z')|$ values. Iterate to height y_0. Each starting point has a finite tree, since every step increases height by ρ. The terminal supremum is finite by periodicity and exponential decay; there is no assumption that the initial global supremum is finite. This proves (2.7).

Write its value as M. If the first $B = B_k$ coefficients vanish, the maximum principle in the Fourier disk, applied from height $y_0/2$, gives for $y \ge y_0$

$$y^\nu |F(x+iy)| \le M(2y/y_0)^\nu e^{-2\pi(B+1)(y-y_0/2)} \le M\, 2^\nu e^{-\pi(B+1)y_0} < M$$

unless $M = 0$. The middle expression decreases for $y \geq y_0$, since $2\pi(B+1) > 2\nu \log 2/y_0 > \nu/y_0$. Taking the terminal supremum forces $M = 0$. Finally, Fourier extraction at height $1/n$ gives $|f_n| n^{(k-1)/2} \leq e^{2\pi} M n^{\nu}$. □

Distinct normalized Euler products are linearly independent: in a minimal finite dependence, apply the coefficient operation $f_n \mapsto f_{pn} + \mathbf{1}_{p\nmid N,\, p|n} f_{n/p}$. Each Euler product is an eigenvector with eigenvalue f_p. Subtracting one eigenvalue times the original dependence forces all f_p to agree; the Euler recurrences then force the sequences to agree. Consequently the union of the finitely many sectors contains only finitely many Euler products, and

$$\#\mathcal{E}(N,k) \leq \sum_e \dim V_k(e). \tag{2.9}$$

This proves the finiteness assertion, including weight two.

3. Reconstruction and a positive operator

3.1. **A bounded extension from a terminal strip.** Use the strict certificate (2.4). Its reconstruction gives constants $\varepsilon > 0$, $\lambda > 1$ and a finite list of rational Möbius maps such that, below height ε, every actual solution is a sum of values at heights at least λy. The coefficients for the normalized section $y^{k/2}F(z)$ have bounded absolute values, independently of k; only their unit-modulus slash phases depend on k. Choose the cell rules once and for all, retaining all integer folds.

Put $d\mu = dx\, dy/y^2$ on the translation cylinder and

$$w_t(y) = \min(y/\varepsilon, 1)^t, \qquad \mathcal{H}_t = L^2((\mathbb{R}/\mathbb{Z}) \times \mathbb{R}_{>0}, w_t^2 d\mu), \qquad \mathcal{G} = L^2((\mathbb{R}/\mathbb{Z}) \times [\varepsilon, \infty), d\mu).$$

Lemma 3.1. *For some fixed $t_0 > 0$, reconstruction defines operators $R_k : \mathcal{G} \to \mathcal{H}_{t_0}$, with norm bounded independently of k, which are the identity on the terminal strip. Every member of $V_k(e)$ is reconstructed from its terminal values. The same statements hold in the regular orbit representation introduced in Section 4.*

Proof. Conjugate the low-height equation by w_t and write it as $u = K_t u + B_t h$. A low-to-low branch with source height y and target height v has the additional factor $(y/v)^t \leq \lambda^{-t}$. For a reconstruction branch containing a Fricke inversion, its determinant-one representative $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has $c^2 = \ell_N(q) \geq N$ for the modulus q of the row used; in particular $c \neq 0$. The possible preimages of a folded target $x + iv$ have height ratios

$$r_n = \frac{y_n}{v} = \frac{1}{c^2((n + x - a/c)^2 + v^2)}.$$

Keeping only ascending branches, $r_n \leq \lambda^{-1}$, and separation of the nearest integer gives

$$\sum_n r_n \leq \lambda^{-1} + 16/c^2, \qquad \sum_n r_n^t \leq \lambda^{-(t-1)}(\lambda^{-1} + 16/c^2).$$

The branch maps preserve μ, so the Schur test applies with the row sums bounded by $C\lambda^{-t}$ (finitely many branches, each with the factor above) and the column sums bounded by $C\lambda^{-(t-1)}$ (the second display). It gives $\|K_t\| \leq C\lambda^{-(t-1/2)}$. On the low-to-terminal part, $v \geq \varepsilon$, and the column sum is bounded using

$$\varepsilon^{-t}|c|^{-2t} v^t \sum_n ((n+\alpha)^2 + v^2)^{-t} \ll_{t,c,\varepsilon} v^{-t} + v^{1-t}.$$

Thus B_t is bounded for each fixed $t \geq 1$.

The other branches are $z \mapsto p(z + d)$ for fixed $p \parallel N$ and $d \in \mathbb{Q}$. On each input cell only finitely many integer folds occur, with output multiplicity at most $p + 1$. Hyperbolic area is preserved, so the folded pullback norm is at most $\sqrt{p+1}$. The low-to-low weight ratio supplies p^{-t}; its norm is

therefore at most $\sqrt{p+1}p^{-t}$. On the low-to-terminal part, $\varepsilon/p \le y < \varepsilon$ and $\varepsilon \le py < p\varepsilon$, so its norm is bounded for fixed t. Its horizontal and output-height tails vanish beyond fixed bounds. The rows at $p^2 \mid N$ contribute no branch. Thus the same contraction and boundedness estimates hold for the whole augmented reconstruction. Choose $t_0 \ge 1$ so that $\|K_{t_0}\| < 1/2$ and $t_0 > s_* + 1/2$, where $s_* = \gamma$ in (2.7). Then

$$R_k h = h \text{ on the terminal strip}, \qquad w_{t_0} R_k h = (I - K_{t_0})^{-1} B_{t_0} h \text{ below it.}$$

Every row of this Neumann series is locally finite because height strictly increases. It is exactly the original reconstruction, not a new boundary condition. The Schur proof works both with counting measure on a rational orbit and with $d\mu$, since Möbius maps preserve hyperbolic area. Finally the growth and cusp bounds of Proposition 2.2 put every actual solution in $\mathcal{H}_{t_0}$. □

Set $t = t_0 + 1$. Let P_k be the orthogonal projection in $\mathcal{H}_t$ onto the normalized periodic holomorphic functions. Its Fourier basis is

$$e_n(z) = \frac{y^{k/2} e^{2\pi i n z}}{\sqrt{\beta_n(t)}}, \quad n \ge 1, \qquad \beta_n(t) = \int_0^\infty y^{k-2} w_t(y)^2 e^{-4\pi n y}\, dy. \tag{3.1}$$

There are no zero or negative Fourier modes, by integrability at infinity.

3.2. **Adding only known equations.** A finite observation operator L_k is a finite stack of known functional-equation or ramified-recurrence residuals, restricted to compact subsets of $\mathbb{H}$. It includes all slash phases and signs. For a fixed observation family its norm is bounded independently of k. Define

$$A_{L,k} = R_k \big[R_k^*(I + L_k^* L_k) R_k\big]^{-1} R_k^*, \qquad Q_{L,k} = P_k A_{L,k} P_k. \tag{3.2}$$

The inverse exists since the terminal identity gives $R_k^* R_k \ge I$. Writing $U_k = R_k (R_k^* R_k)^{-1/2}$ gives

$$A_{L,k} = U_k (I + (L_k U_k)^* (L_k U_k))^{-1} U_k^*.$$

It is a positive contraction. Its holomorphic one-eigenspace contains $V_k(e)$, whence

$$\dim V_k(e) \le \operatorname{tr} Q_{L,k}. \tag{3.3}$$

No unproved individual modularity relation is inserted into L_k.

For later use there is a common summable majorant. The bound into $\mathcal{H}_{t_0}$, applied to the adjoint pairing, gives

$$0 \le Q_{L,k} \le P_k R_k R_k^* P_k \le C\, \operatorname{diag}_{n\ge1} \frac{\beta_n(t+1)}{\beta_n(t)} \le C' \operatorname{diag}_{n\ge1} \min(1, (k/n)^2). \tag{3.4}$$

For completeness, put $a = k + 2t - 1$. If $4\pi n\varepsilon \ge 2a$, the lower incomplete gamma integral in $\beta_n(t)$ is at least half the full gamma integral, by Markov's inequality for a gamma variable of mean a. Bounding the numerator by its uncapped integral gives $\beta_n(t+1)/\beta_n(t) \le 2a(a+1)/(4\pi n\varepsilon)^2$. In particular all the traces in (3.3) are finite and $O_N(k)$.

4. The regular trace and its holomorphic realization

4.1. **Counting coordinates and the rooted trace.** Let $G_\mathbb{Q} = \mathrm{PGL}_2^+(\mathbb{Q})$ act on $\mathbb{H}$. Identify $\Sigma = \langle T_1 \rangle \backslash \mathbb{H}$ with the strip $0 \le \Re z < 1$, carrying μ, and write $x \sim x'$ when $x' \in G_\mathbb{Q} x$. This relation $\mathcal{R} \subset \Sigma \times \Sigma$ has countable classes. For $g \in G_\mathbb{Q}$ the map $z \mapsto gz$, restricted to the points of the strip with image in the strip, is a partial Borel isomorphism of Σ preserving μ, and $\mathcal{R}$ is the union of these countably many graphs. Cutting $\mathcal{R}$ into disjoint measurable pieces of the graphs and changing variables on each gives, for measurable $f \ge 0$ on $\mathcal{R}$,

$$\int_\Sigma \sum_{x' \sim x} f(x, x')\, d\mu(x) = \int_\Sigma \sum_{x' \sim x} f(x', x)\, d\mu(x). \tag{4.1}$$

Let $\mathcal{M}$ be the von Neumann algebra of the countable measure-preserving relation $\mathcal{R}$ [3]. Its elements are bounded families of operators on ℓ^2 of the classes (counting measure) whose matrix coefficients $\langle \delta_x, B\delta_{x'} \rangle$ are measurable on $\mathcal{R}$, families agreeing on almost every class being identified. A measurable kernel on $\mathcal{R}$ with bounded absolute row and column sums defines an element of $\mathcal{M}$, a strong limit of kernels with boundedly many entries per row and column. This covers rational branches with cell multipliers and the regular versions of K_t, B_t in Lemma 3.1, whose columns are infinite.

For a positive $B \in \mathcal{M}$, its rooted trace is

$$\tau(B) = \int_\Sigma \langle \delta_{[z]}, B\delta_{[z]} \rangle \, d\mu(z). \tag{4.2}$$

This is the canonical weight of $\mathcal{M}$ [3]: faithful, normal and semifinite, and a trace because μ is invariant. The trace identity needs no finiteness of rows or columns: for $B \in \mathcal{M}$ and $f(x, x') = |\langle \delta_x, B\delta_{x'} \rangle|^2$, the two sides of (4.1) are $\tau(BB^*) = \int_\Sigma \|B^*\delta_x\|^2 \, d\mu$ and $\tau(B^*B) = \int_\Sigma \|B\delta_x\|^2 \, d\mu$. Labels in the output of an observation operator are counted separately.

None of this uses freeness. Off the null set of fixed points of the countably many nonidentity elements of $G_\mathbb{Q}$, orbits are free; this is used to remove the weight-k phases by a gauge (below), to decide which branch words return to the root (Proposition 4.1, Lemma 5.1), and for the coset bijections and rooted diagonals of Lemma 6.1 and Section 6.3. The identity $I_\mathcal{G}$ of the terminal corner is multiplication by $\mathbf{1}_{y \geq \varepsilon}$. In particular

$$\tau(I_\mathcal{G}) = \int_\varepsilon^\infty \frac{dy}{y^2} = \varepsilon^{-1}.$$

The usual dimension identities for closed module kernels and ranges follow from polar decomposition in this finite terminal corner.

The weight-k unit phases can be removed on a free orbit by the slash cocycle. A choice of root gives a diagonal unitary gauge; closed-path trace coefficients are independent of that choice. The remaining coefficients are the fixed signs, cell multipliers and capped-weight ratios. Thus the regular version A_L of (3.2) has a trace independent of k. This does not put actual holomorphic functions in the counting Hilbert spaces.

4.2. **The fixed-observation trace comparison.**

Proposition 4.1. *For every fixed finite compact observation family L,*

$$\lim_{k \to \infty} \frac{\operatorname{tr} Q_{L,k}}{k} = \frac{\tau(A_L)}{4\pi}. \tag{4.3}$$

All cutoffs, observations and their scalar multipliers are fixed before taking the weight limit.

Proof. We supply the approximation steps, including the weighted projection. First cut the output of reconstruction below height $Y < \varepsilon$. Since $t = t_0 + 1$,

$$\|R_k - \mathbf{1}_{y \geq Y} R_k\|_{\mathcal{G} \to \mathcal{H}_t} \leq CY/\varepsilon.$$

Take Y below all source *and image* heights in L. Then the observation of the removed part is zero, and the change in the positive inverse normalization is $O((Y/\varepsilon)^2)$. The resulting positive contractions differ in norm by $O(Y/\varepsilon)$ and retain (3.4). To pass from norm error δ to normalized trace error, split the Fourier sum at Ak: its error is $O(kA\delta + k/A)$. This tends to zero after division by k, first as $\delta \to 0$ and then $A \to \infty$.

On the regular side norm convergence does not suffice, as τ is only semifinite below the terminal strip. The trace identity for $U(I + (LU)^*(LU))^{-1/2}$ gives

$$\tau(A_L) = \tau\big((I + (LU)^*(LU))^{-1}\big),$$

a trace in the finite terminal corner, where $|\tau(B)| \le \varepsilon^{-1}\|B\|$. The same holds after the cutoff, with $R^Y = \mathbf{1}_{y\ge Y}R$ and $U^Y = R^Y(R^{Y*}R^Y)^{-1/2}$, R being the regular reconstruction. Now $LR^Y = LR$, while $R^{Y*}R^Y$ and R^*R are at least I and differ in norm by $O((Y/\varepsilon)^2)$. Hence $LU^Y \to LU$ in norm, and the cut-off regular traces converge to $\tau(A_L)$ as $Y \to 0$.

At fixed Y there are finitely many reconstruction paths: their depth is at most $1+\lceil \log(\varepsilon/Y)/\log\lambda\rceil$. For every fixed branch with $c \neq 0$,

$$\Im(gz) \le (c^2Y)^{-1}, \qquad |\Re(gz)| \le |a/c| + (c^2Y)^{-1}.$$

For an affine branch the finite-fold bound above applies and its terminal output height is at most $p\varepsilon$. Thus there are only finitely many integer folds and a fixed upper bound on all terminal heights sampled from the retained low region. Above a sufficiently large fixed H, reconstruction is just terminal inclusion and the observations vanish. That block of $A_{L,k}$ is the identity. The eigenvalues of its holomorphic compression satisfy

$$g_n(H) \le \min(1, 2^{k+2t-1}e^{-2\pi nH}),$$

by rescaling the norm integral in (3.1). Hence

$$\operatorname{tr}(P_k\mathbf{1}_{y>H}P_k) \le \frac{(k+2t-1)\log 2}{2\pi H} + \frac{1}{1-e^{-2\pi H}}. \tag{4.4}$$

The regular trace of that block is $1/H$. Both normalized tails tend to zero as $H \to \infty$.

On the retained compact region the reconstruction is a finite rational branch sum. Its normalization inverse has spectrum in a fixed interval $[1, C_L]$ and can be uniformly approximated by polynomials. An inverse error δ costs $O(\delta k)$ in the holomorphic trace, by the factorization through P_kR_k and (3.4), and $O(\delta)$ in the finite regular trace. It therefore suffices to treat fixed finite branch words.

Here is the needed comparison of weighted and unweighted projections. Conjugate P_k by multiplication by w_t to the unweighted ambient $L^2(d\mu)$. In Fourier mode n its unit radial vector differs from the unweighted vector by angle $O_t(k^{-1/2})$, uniformly in n. Indeed, for a gamma variable X of shape $K = k-1$, set $y_n = K/(4\pi n)$ and $r = w_t(y_nX/K)/w_t(y_n)$. Then

$$|r-1| \le |(X/K)^t - 1|, \qquad \mathbb{E}(r-1)^2 = O_t(K^{-1}),$$

using $\mathbb{E}(X/K)^u = \Gamma(K+u)/(\Gamma(K)K^u)$. The squared cosine of the angle is $(\mathbb{E}r)^2/\mathbb{E}r^2$. Rotate the two radial vectors in each mode. The resulting ambient unitary is $I + O_t(k^{-1/2})$ in norm.

Let P_w and P_0 be the conjugated weighted and the unweighted projections, χ the indicator of the retained region $Y \le y \le H$, and $\|\cdot\|_1$, $\|\cdot\|_2$ the trace and Hilbert–Schmidt norms; every fixed word W in the retained region satisfies $W = \chi W\chi$. The proof of (4.4), with Y for H (and $t = 0$ for P_0), gives $\|\chi P_w\|_2^2 + \|P_0\chi\|_2^2 = O(k)$. Also $\|(P_w - P_0)\chi\|_2 = O(1)$: by the angle bound the modes $n \le Ck$ contribute $O(k\cdot k^{-1})$ to its square, and each mode $n > Ck$ at most $4\cdot 2^{k+2t-1}e^{-2\pi nY}$, with exponentially small sum once $C > \log 2/(2\pi Y)$. Since $P_w - P_0 = P_w(P_w - P_0) + (P_w - P_0)P_0$,

$$\|\chi(P_w - P_0)\chi\|_1 \le \|\chi P_w\|_2\,\|(P_w - P_0)\chi\|_2 + \|\chi(P_w - P_0)\|_2\,\|P_0\chi\|_2 = O(\sqrt{k}).$$

As $\operatorname{tr}(P_wWP_w) = \operatorname{tr}(\chi P_w\chi W)$, and likewise for P_0, replacing the weighted projection by the unweighted one changes each fixed composed trace by at most $\|\chi(P_w - P_0)\chi\|_1\|W\| = O(\sqrt{k})$. The conjugated branch coefficients still contain $w_t(y)/w_t(\Im(gz))$.

For the unweighted holomorphic projection the normalized kernel on $\mathbb{H}$ is

$$\frac{k-1}{4\pi}\left(\frac{2i\sqrt{yv}}{z-\overline{u}}\right)^k, \qquad z = x+iy, \quad u = \xi + iv.$$

Periodizing in the real direction gives the cylinder kernel. In a trace of a fixed branch word, a term of order k occurs exactly when the composed rational map returns the root modulo an integral translation. Its coefficient is the integral of the corresponding rooted diagonal, multiplied by $1/(4\pi)$.

For every other word, the modulus of the displayed kernel divided by $(k-1)/(4\pi)$ tends to zero off the fixed-point set, which has area zero. On compact sets it is bounded by an integrable constant; the periodization tails decay uniformly. Dominated convergence makes this contribution $o(k)$. No regularity of the cell rules enters. A fixed word is a finite sum of terms $\phi \mapsto m(z)\phi(\gamma z)$, with γ a composed rational map and m bounded, measurable and supported where z and γz lie in the retained region. As the projection occurs only once, the trace of such a term is exactly $\int m(z)K(\gamma z, z)\, d\mu(z)$, K being the cylinder kernel. The integral converges absolutely: $|K(\gamma z, z)|^2 \le K(\gamma z, \gamma z)K(z, z)$, and the diagonal of K is integrable over the retained region. This proves the required identity-return formula for each finite approximant.

Its rooted coefficients are precisely (4.2). Remove polynomial errors and height cutoffs *after* the weight limit, using the preceding estimates and normality of the regular trace. This proves (4.3). □

4.3. **Uniform cutoff and finite-branch estimates.**

Lemma 4.2 (A cutoff independent of the penalty)**.** *Let L be a finite compact observation stack, $s \ge 0$, and $R^Y = \mathbf{1}_{y \ge Y} R$, where $0 < Y < \varepsilon$ is below every source and image height in L. Define A_s by* (3.2) *with observation sL, and define A_s^Y by replacing R by R^Y. Then, uniformly in s and in such L,*

$$\|A_{s,k} - A_{s,k}^Y\| \ll_N Y, \qquad k^{-1}\big|\mathrm{tr}(P_k(A_{s,k} - A_{s,k}^Y)P_k)\big| \ll_N \sqrt{Y}, \qquad |\tau(A_s) - \tau(A_s^Y)| \ll_N Y^2.$$

Proof. Put $G = R^*R$, $G_Y = R^{Y*}R^Y$, and

$$B_s = G + s^2 R^* L^* L R, \qquad B_{s,Y} = G_Y + s^2 R^{Y*} L^* L R^Y.$$

Here and below the same identities hold with the weight-k phases. The reconstruction bound gives $\|R - R^Y\| \ll_N Y$ and, since the removed and retained outputs are orthogonal, $\|G - G_Y\| \ll_N Y^2$. Moreover $LR^Y = LR$, so $B_s - B_{s,Y} = G - G_Y$ independently of s. Both positive operators are at least I on the terminal corner. The resolvent identity therefore gives $\|B_s^{-1} - B_{s,Y}^{-1}\| \ll_N Y^2$. Expanding $RB_s^{-1}R^* - R^Y B_{s,Y}^{-1} R^{Y*}$ proves the norm bound. Both positive compressions retain (3.4). Splitting the Fourier trace at Ak bounds the normalized trace difference by $O_N(AY + A^{-1})$; take $A = Y^{-1/2}$ (and adjust the constant for Y bounded away from zero). Finally, on the regular side cyclicity moves both traces into the finite terminal corner:

$$\tau(A_s) = \tau(GB_s^{-1}), \qquad \tau(A_s^Y) = \tau(G_Y B_{s,Y}^{-1}).$$

The two products differ in norm by $O_N(Y^2)$, and the corner has trace ε^{-1}. This proves the last assertion. □

Lemma 4.3 (A quantitative finite-branch comparison)**.** *Fix $t \ge 0$ and the capped weight w_t used above. Let $k \ge 6$ be even, $0 < Y \le 1 \le H$, and $B \ge 1$. Put*

$$K_{Y,H} = \{x + iy : 0 \le x < 1,\ Y \le y \le H\}, \qquad \chi = \mathbf{1}_{K_{Y,H}}.$$

In the unweighted realization $L^2(\Sigma, d\mu)$, let

$$(W_k\phi)(z) = \sum_{\nu=1}^{S} a_\nu(z) j_k(g_\nu, z)\phi(g_\nu z), \qquad j_k(g, z) = \left(\frac{cz+d}{|cz+d|}\right)^{-k} \quad \left(g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right).$$

Here each g_ν has a primitive integral representative of positive determinant with entries of absolute value at most B; the measurable coefficient a_ν, independent of k, has absolute value at most M and is supported on a measurable set $E_\nu \subset K_{Y,H}$ with $g_\nu E_\nu \subset K_{Y,H}$. In particular, all integer folds are

included in the branch list. Let $P_{w,k}$ be the projection P_k conjugated by multiplication by w_t, and let W be the corresponding regular operator after removal of the slash phases. Then

$$\left|\frac{\operatorname{tr}(P_{w,k}W_kP_{w,k})}{k}-\frac{\tau(W)}{4\pi}\right|\ll_t \frac{SMBH^2}{Y^2\sqrt{k}}. \tag{4.5}$$

The implicit constant is independent of the measurable sets and coefficients, and of S, M, B, Y, H, k. In particular it can be replaced by $C_tSM(1+B+H+Y^{-1})^5k^{-1/2}$.

Proof. Each individual branch has operator norm at most M, since a real Möbius transformation preserves μ, and hence $\|W_k\|\le SM$ and $W_k=\chi W_k\chi$. We first make the comparison of weighted projections uniform in Y. Write $P_{0,k}$ for the unweighted periodic holomorphic projection. The gamma-moment argument in the proof of Proposition 4.1 shows that, in every positive Fourier mode, the squared angle between the two radial lines is $O_t(k^{-1})$, uniformly in the mode. This also follows directly by putting $K=k-1$, taking X gamma distributed with shape K, and using

$$\left|\frac{w_t(yX/K)}{w_t(y)}-1\right|\le |(X/K)^t-1|,\qquad \mathbb{E}|(X/K)^t-1|^2\ll_t K^{-1}.$$

The second assertion follows by expanding the square and applying $\mathbb{E}(X/K)^u=\Gamma(K+u)/(\Gamma(K)K^u)$ for $u=t,2t$. For sufficiently large k the denominator in the squared-cosine formula is bounded below; the finitely many smaller $k\ge 6$ are absorbed into the constant.

For either radial line, its squared mass above Y in Fourier mode n is at most

$$\min\{1,2^{k+2t-1}e^{-2\pi nY}\};$$

for the unweighted line one can take $t=0$. Indeed, split $e^{-4\pi ny}=e^{-2\pi ny}e^{-2\pi ny}$, use $y\ge Y$ in one factor, and change variables $y=2u$ in the remaining integral, using $w_t(2u)\le 2^tw_t(u)$. Consequently

$$\|\chi P_{w,k}\|_2^2+\|P_{0,k}\chi\|_2^2\ll_t k/Y.$$

In the first $\lceil C_tk/Y\rceil$ modes the squared Hilbert–Schmidt norm of the difference of the two rank-one projections is $O_t(k^{-1})$. In the remaining modes its localized squared Hilbert–Schmidt norm is bounded by twice the sum of the two radial tail masses. Taking C_t sufficiently large and summing the geometric tail gives

$$\|(P_{w,k}-P_{0,k})\chi\|_2^2\ll_t Y^{-1}.$$

Here $1/(1-e^{-2\pi Y})\ll Y^{-1}$ because $Y\le 1$. The identity $P_{w,k}-P_{0,k}=P_{w,k}(P_{w,k}-P_{0,k})+(P_{w,k}-P_{0,k})P_{0,k}$ now gives

$$\|\chi(P_{w,k}-P_{0,k})\chi\|_1\ll_t \sqrt{k}/Y. \tag{4.6}$$

It follows that replacing $P_{w,k}$ by $P_{0,k}$ in the trace costs $O_t(SM\sqrt{k}/Y)$.

We next estimate a single unweighted branch. The normalized kernel on $\mathbb{H}$ is

$$K_k^{\mathbb{H}}(u,z)=\frac{k-1}{4\pi}\left(\frac{2i\sqrt{\Im u\Im z}}{u-\overline{z}}\right)^k,$$

and the cylinder kernel is its sum over integral translates of the first variable. The trace of the branch is therefore the integral of $a(z)j_k(g,z)K_k^{\Sigma}(gz,z)$ over its support. This trace identity is justified, for example, by expansion in an orthonormal basis and Cauchy–Schwarz: both localized diagonal integrals are finite, and changing variables by g preserves μ.

Choose the stipulated integral representative $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ of g, and write $D=\det A$, so that $1\le D\le 2B^2$. For the translate T_ng use the integral representative

$$A_n=T_nA=\begin{pmatrix}a+nc & b+nd\\ c & d\end{pmatrix}.$$

Its entries can grow with n, but its determinant remains exactly D. Define its integral displacement polynomial

$$p_n(z) = cz^2 + (d - a - nc)z - (b + nd).$$

A direct calculation gives

$$\frac{|K_k^{\mathbb{H}}(T_n gz, z)|}{(k-1)/(4\pi)} = \left(1 + \frac{|p_n(z)|^2}{4y^2 D}\right)^{-k/2}. \tag{4.7}$$

If p_n is nonconstant, its sublevel sets in the whole complex plane have Euclidean area

$$\operatorname{area}\{|p_n(z)| \le r\} \le C(r + r^2) \qquad (r > 0), \tag{4.8}$$

with an absolute constant. For a linear polynomial its leading coefficient is a nonzero integer, so the set is a disk of area at most πr^2. For a quadratic, factoring it over $\mathbb{C}$ shows that at least one factor has absolute value at most $\sqrt{r}$, because the absolute value of its integral leading coefficient is at least one. The set is therefore contained in two disks of total area at most $2\pi r$.

Put $A_* = \sqrt{8}BH$. Since $y \le H$, the right side of (4.7) is at most $f(|p_n(z)|)$, where $f(r) = (1 + r^2/A_*^2)^{-k/2}$. Layer integration using (4.8), followed by integration by parts, gives

$$\int_{\mathbb{C}} f(|p_n(z)|)\, dx\, dy \le C \int_0^\infty f(r)(1 + 2r)\, dr \ll \frac{BH}{\sqrt{k}} + \frac{B^2H^2}{k}.$$

The last estimate holds uniformly for $k \ge 6$; the two integrals are an elementary beta integral and $A_*^2/(k-2)$, respectively. On $K_{Y,H}$, $d\mu \le Y^{-2} dx\, dy$. Thus every such translate has normalized absolute integral at most

$$CY^{-2}\left(\frac{BH}{\sqrt{k}} + \frac{B^2H^2}{k}\right). \tag{4.9}$$

If p_n is a nonzero constant, its absolute value is at least one. Since $\mu(K_{Y,H}) \le Y^{-1}$, its normalized integral is at most

$$Y^{-1}(1 + A_*^{-2})^{-k/2} \ll Y^{-1}BH/\sqrt{k},$$

which is also bounded by (4.9).

It remains to sum the translates. Since both z and gz are in the chosen strip and their heights are at most H, the normalized kernel for $|n| > 4H + 4$ is at most

$$\left(1 + \frac{(|n| - 1)^2}{4H^2}\right)^{-k/2}.$$

Summing this decreasing tail by comparison with its integral gives $O(He^{-ck})$ for an absolute $c > 0$. Its integrated contribution is $O(HY^{-1}e^{-ck})$. There are $O(H)$ remaining translates, to each of which (4.9) applies unless $p_n = 0$.

The condition $p_n = 0$ says exactly that $T_n g$ is a scalar matrix, that is, the branch returns the root modulo an integral translation. On a branch folded into the chosen strip this can have nonzero support only for $g = I$ in $\mathrm{PGL}_2^+(\mathbb{Q})$ and $n = 0$. The slash phase is then one because k is even, and its contribution is exactly $(k-1)/(4\pi)$ times the corresponding regular rooted diagonal integral. All other fixed-point sets have measure zero. Summing over the S branches, and accounting for the replacement of $k - 1$ by k, proves

$$\left|\frac{\operatorname{tr}(P_{0,k} W_k P_{0,k})}{k} - \frac{\tau(W)}{4\pi}\right| \ll SM\left\{\frac{BH^2}{Y^2\sqrt{k}} + \frac{B^2H^3}{Y^2 k} + \frac{H}{Y}e^{-ck} + \frac{1}{kY}\right\}.$$

When $k \ge B^2H^2$, the first term bounds all the others. For completeness, when $6 \le k < B^2H^2$ use instead the diagonal bound

$$K_k^{\Sigma}(z, z) \ll k(1 + H/\sqrt{k}) \qquad (z \in K_{Y,H}),$$

obtained by summing $(1+n^2/(4y^2))^{-k/2}$ and comparing with its integral. Kernel Cauchy–Schwarz and the measure-preserving change of variables give a normalized trace bound $CSM(1+H/\sqrt{k})/Y$, whereas $|\tau(W)| \le SM/Y$. Both are at most $CSMBH^2/(Y^2\sqrt{k})$ in this range. Finally (4.6) is absorbed by the same bound. This proves (4.5). □

5. Fresh primes remove the regular excess

5.1. **One-prime rank loss.** Let Q be the modulus of the fixed certificate; the reconstruction branches are $T_{x'}W_{\ell_N(q')}T_x$ with $q' \mid Q$ and $Qx, Qx' \in \mathbb{Z}$, together with the finitely many ramified dilation branches. Enlarge Q to include every fixed denominator if necessary. Let P be the terminal kernel projection after finitely many previously imposed equation families; global equations mean intersections of their compact observations. Choose a prime $q \ge 3$ with $q \nmid NQ$, dividing no modulus q' of an earlier equation (2.1), and put $H_q = \mathrm{PGL}_2^+(\mathbb{Q}) \cap \mathrm{PGL}_2(\mathbb{Z}_q)$. Translations by rationals with denominator prime to q, the matrices $W_{\ell_N(q')}$ with $q \nmid Nq'$, and the matrices $\begin{pmatrix} 1 & a \\ 0 & p \end{pmatrix}$ of determinant $p \mid N$ by which the operators $\mathcal{U}_p$ of Section 6.1 observe the ramified recurrences (2.2) have entries in $\mathbb{Z}_q$ and determinant in $\mathbb{Z}_q^\times$. The added dilation matrices have determinant $p \parallel N$, also a q-adic unit. Hence every branch of the reconstruction and of the earlier equations lies in H_q.

Call an operator family *q-integral* if its matrix coefficient between two points, or translation classes, of an orbit vanishes unless they lie in one H_q-orbit (meaningful for classes, as $T_1 \in H_q$). These are the families commuting with the projections onto the H_q-orbits, hence closed under products, adjoints, strong limits and Borel functional calculus. Diagonal multipliers and branches in H_q are q-integral; so therefore are the regular reconstruction, U, the earlier observations, their kernel projections and P.

For $1 \le b < q$, choose $1 \le a(b) < q$ with $Na(b)b \equiv -1 \pmod q$. The matrices

$$g_b = T_{a(b)/q}W_{Nq^2}T_{-b/q} \sim \begin{pmatrix} Na(b) & -(Na(b)b+1)/q \\ Nq & -Nb \end{pmatrix} \tag{5.1}$$

have determinant N in the displayed representatives. They lie in $\Gamma_0(N)W_N$ and are integral at q.

The defects $D_bF = wF|g_b - F$ are not periodic, so their regular versions take values in ℓ^2 of full orbits: we work in the algebra of the orbit relation of $G_\mathbb{Q}$ on $(\mathbb{H}, \mu)$, built as in Section 4, of which $\mathcal{M}$ is the corner cut by the strip, with the restricted trace. Since $b \mapsto a(b)$ permutes the units and $g_bT_{b/q} = T_{a(b)/q}W_{Nq^2}$, the prime-$q$ equation (2.1) reads

$$\sum_{b=1}^{q-1} (D_bF)(\zeta + b/q) = 0 \qquad (\zeta \in \mathbb{H});$$

it does not assert that the defects vanish individually.

Fix a compact $\Omega \subset \mathbb{H}$ and let $(\theta_s\psi)(\zeta) = \psi(\zeta + s)$ on full orbits. Let $X = (X_b)_b$ with $X_b = \mathbf{1}_{\Omega+b/q}D_bUP$, U being the regular version of U_k, and $v\psi = (\theta^*_{r/q}\psi)_r$. Then $v^*X = \mathbf{1}_\Omega \sum_b \theta_{b/q}D_bUP$ is the known equation on Ω, $v^*v = (q-1)I$, and $C = vv^*$, with entries $C_{rb} = \theta_{(b-r)/q}$, satisfies $C^2 = (q-1)C$. So $J = C/(q-1)$ is an orthogonal projection and $\ker CX = \ker v^*X$. Put $H = X^*X$, $R = \operatorname{supp} H$, and let $V = (V_b)_b$ be the polar isometry: $X = VH^{1/2}$, $V^*V = R$.

Lemma 5.1. *Let q be as above, and let P', P'' be the projections onto the kernels within P of the averaged equation v^*X and of the individual defects X. Then*

$$\tau(P) - \tau(P') \ge \frac{\tau(P) - \tau(P'')}{q-1}. \tag{5.2}$$

The same holds for the global kernels, the intersections over all Ω.

Proof. As $g_b \in H_q$, each X_b is q-integral; so are H and the strong limits $V_b = \lim_{\eta\to 0^+} X_b(H+\eta)^{-1/2}$. Let $x = [z]$ be a terminal root on a free orbit. Then $V_b\delta_x$ is supported in $H_q z$ and $\theta_s V_b\delta_x$ in $T_{-s}H_q z$. For $r \neq b$ and $s = (b-r)/q$ we have $s \notin \mathbb{Z}_q$, so $T_{-s} \notin H_q$, and these sets are disjoint: $T_{-s}hz = h'z$ with $h, h' \in H_q$ would give $T_{-s} = h'h^{-1}$ by freeness. Hence $\langle V_r\delta_x, \theta_{(b-r)/q}V_b\delta_x\rangle = 0$ for $r \neq b$, and

$$\langle \delta_x, V^*CV\delta_x\rangle = \sum_b \|V_b\delta_x\|^2 = \langle \delta_x, R\delta_x\rangle.$$

Integrating over these roots, a conull set, gives $\tau(E) = \tau(R)/(q-1)$ for $E = V^*JV$. As $0 \le E \le R$, the support S_E of E has $\tau(S_E) \ge \tau(E)$. Now $(CX)^*CX = (q-1)^2H^{1/2}EH^{1/2}$ has support $P - P'$, the right support of $E^{1/2}H^{1/2}$. Its left support is S_E, since the range of $H^{1/2}$ is dense in that of R and $E^{1/2}R = E^{1/2}$. Left and right supports have equal trace, by polar decomposition in the finite terminal corner. As $R = P - P''$,

$$\tau(P) - \tau(P') = \tau(S_E) \ge \frac{\tau(R)}{q-1} = \frac{\tau(P) - \tau(P'')}{q-1}.$$

Finally let Ω increase to $\mathbb{H}$: P' and P'' decrease to the global kernel projections, and normality of the finite terminal trace passes the inequality to the limit. □

Lemma 5.2 (Fresh-prime spectral loss)**.** *In the setting of Lemma 5.1, put $Y_q = v^*X$ and $H = X^*X$. For every $h > 0$,*

$$\tau\big(\mathbf{1}_{[h/4,\infty)}(Y_q^*Y_q)\big) \ge \frac{1}{2(q-1)}\tau\big(\mathbf{1}_{[h,\infty)}(H)\big). \tag{5.3}$$

More generally the same assertion holds for any bounded q-integral column X with the indicated translated output components and a finite-trace source corner. For every fixed $0 < \theta < 1$, the sharper version is

$$\tau\big(\mathbf{1}_{[\theta h/2,\infty)}(Y_q^*Y_q)\big) \ge \frac{1-\theta}{q-1}\tau\big(\mathbf{1}_{[h,\infty)}(H)\big). \tag{5.4}$$

Proof. Write $n = q-1$, $X = VH^{1/2}$, and $R_h = \mathbf{1}_{[h,\infty)}(H)$. The fresh-prime orthogonality argument applied to VR_h gives

$$E = R_hV^*JVR_h, \qquad 0 \le E \le R_h, \qquad \tau(E) = \tau(R_h)/n.$$

For $F = \mathbf{1}_{[\theta/n,\infty)}(E)$, considered in the R_h corner,

$$\tau(R_h)/n = \tau(E) \le \tau(F) + \theta\tau(R_h)/n,$$

so $\tau(F) \ge (1-\theta)\tau(R_h)/n$. The bounded invertible map $H^{-1/2}$ on R_h sends $\operatorname{Ran} F$ onto a closed submodule of the same dimension. If $u = H^{-1/2}f$ with $f \in \operatorname{Ran} F$, then

$$\langle Y_q^*Y_qu, u\rangle = n\langle Ef, f\rangle \ge \theta\|f\|^2 \ge \theta h\|u\|^2.$$

The spectral dimension principle now bounds the dimension of this submodule by that of the spectral projection $\mathbf{1}_{[\theta h/2,\infty)}(Y_q^*Y_q)$. This proves (5.4); taking $\theta = 1/2$ gives (5.3). □

6. Ramified recurrences and the local new trace

6.1. **The two old directions.** On $\Gamma_0(N)$-invariant vectors use

$$\mathcal{U}_p = \sum_{a \bmod p} |_k \begin{pmatrix} 1 & a \\ 0 & p \end{pmatrix}, \qquad D_p = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}.$$

With the normalized Fourier coefficients fixed in Section 1, Fourier summation shows that (2.2) is exactly $\mathcal{U}_pF = \delta_pF$ for $p \parallel N$ and $\mathcal{U}_pF = 0$ for $p^2 \mid N$; in the regular module the observation is this branch equation itself.

For $p \parallel N$, put $N = pN_0$. The identity

$$\frac{1}{p} T_{a/p} W_{N_0 p^2} T_{-b/p} = \begin{pmatrix} N_0 a & -(N_0 ab + 1)/p \\ N & -N_0 b \end{pmatrix}, \qquad N_0 ab \equiv -1 \pmod{p},$$

represents the Atkin–Lehner operator W_{N_0}. Thus on $\Gamma_0(N)$ invariants the ramified equation gives $F|W_{N_0} R_p = -w\delta_p F|R_p$. On period-one functions

$$R_p^2 = (p-2)R_p + (p-1)I,$$

so R_p is invertible, and consequently

$$F|W_p = -\delta_p F. \tag{6.1}$$

Trace to level N/p is $I + W_p \mathcal{U}_p$: if $W_p = \begin{pmatrix} pA & B \\ NC & pD \end{pmatrix}$, the p matrices $W_p \begin{pmatrix} 1 & a \\ 0 & p \end{pmatrix} /p$, together with I, give the coset representatives. Equations (6.1) and (2.2) annihilate this trace, since $\delta_p^2 = 1$. The involution exchanges the two old degeneracy ranges, so both old traces vanish.

If $p^2 \mid N$, the conjugate group

$$D_p \Gamma_0(N) D_p^{-1} = \left\{ \begin{pmatrix} a & pb \\ (N/p)c & d \end{pmatrix} \right\}$$

has index p in $\Gamma_0(N/p)$, with representatives T_a, $0 \le a < p$. Trace of $F|D_p^{-1}$ is precisely $\mathcal{U}_p F$. Thus one old trace vanishes. The identity $W_N = W_{N/p} D_p$ and the global Fricke eigenrelation exchange the old directions and annihilate the other trace too. These are statements about the regular quotient modules; for classical cusp forms they are the familiar orthogonality characterization of newness.

6.2. **Removing positive coordinate weights.** We justify using unweighted incidence equations in this calculation. In a finite-trace source module let D be an equivariant finite-row matrix and let b be a positive equivariant diagonal multiplier, finite and nonzero almost everywhere. Define kernels by the coordinate equations $\mathsf{D}x = 0$ and $\mathsf{D}(bx) = 0$, without requiring bx globally square summable. They have the same trace dimension. Indeed, with $E_m = \mathbf{1}_{[1/m,m]}(b)$,

$$\tau(K \cap E_m) \ge \tau(K) - \tau(I - E_m).$$

On this intersection b^{-1} is bounded and bounded below and maps into the weighted kernel. Apply the reverse argument with b, then let $m \to \infty$. Polar decomposition and normality of finite trace prove equality.

Write $\Gamma = \Gamma_0(N)$ and

$$s_\Gamma(z) = \sum_{\gamma \in \langle T_1 \rangle \backslash \Gamma} w_t(\Im \gamma z)^2.$$

A coset is determined by the lower row $\pm(c, d)$ of γ. The term with $c = 0$ is $w_t(\Im z)^2 > 0$, and the others are at most $\varepsilon^{-2t} (\Im z)^{2t} |cz + d|^{-4t}$. Since $2t > 1$, the function s_Γ is positive, finite and Γ-invariant. Let $\mathcal{M}_\Gamma$ be the algebra of Section 4 formed with $\Gamma\backslash\mathbb{H}$ in place of $\Sigma = \langle T_1 \rangle \backslash \mathbb{H}$. It acts on $\ell^2(\Gamma\backslash\mathcal{O})$ for free orbits $\mathcal{O}$, and its trace

$$\tau_\Gamma(B) = \int_{\Gamma\backslash\mathbb{H}} \langle \delta_x, B\delta_x \rangle \, d\mu(x)$$

is finite, with $\tau_\Gamma(I) = \mu(\Gamma\backslash\mathbb{H})$.

Lemma 6.1. *In counting coordinates $u = w_t \psi$ on a free orbit $\mathcal{O}$, define*

$$\iota_\Gamma \colon \ell^2(\Gamma\backslash\mathcal{O}) \to \ell^2(\langle T_1 \rangle \backslash \mathcal{O}), \qquad (\iota_\Gamma \phi)([\zeta]) = w_t(\Im \zeta)\, s_\Gamma(\zeta)^{-1/2} \phi(\Gamma\zeta).$$

Then ι_Γ is an isometry. Its range consists of the square-summable $w_t\psi$ with ψ invariant under Γ, where $\psi = s_\Gamma^{-1/2}\phi$. For every positive $B \in \mathcal{M}_\Gamma$,

$$\tau(\iota_\Gamma B \iota_\Gamma^*) = \tau_\Gamma(B).$$

In particular $\iota_\Gamma\iota_\Gamma^*$ has rooted diagonal $w_t(\Im\zeta)^2/s_\Gamma(\zeta)$ and trace $\mu(\Gamma\backslash\mathbb{H})$.

Proof. On a free orbit, $\langle T_1\rangle\gamma \mapsto [\gamma\zeta]$ is a bijection from $\langle T_1\rangle\backslash\Gamma$ onto the translation classes in $\Gamma\zeta$. Hence

$$\|\iota_\Gamma\phi\|^2 = \sum_{\Gamma\zeta} \frac{|\phi(\Gamma\zeta)|^2}{s_\Gamma(\zeta)} \sum_\gamma w_t(\Im\gamma\zeta)^2 = \|\phi\|^2,$$

and the range is as stated. Since $\iota_\Gamma^*\delta_{[\zeta]} = w_t(\Im\zeta)s_\Gamma(\zeta)^{-1/2}\delta_{\Gamma\zeta}$,

$$\langle \delta_{[\zeta]}, \iota_\Gamma B\iota_\Gamma^*\delta_{[\zeta]}\rangle = \frac{w_t(\Im\zeta)^2}{s_\Gamma(\zeta)}\, b(\Gamma\zeta), \qquad b(x) = \langle \delta_x, B\delta_x\rangle.$$

For measurable $f \ge 0$ one has $\int_{\langle T_1\rangle\backslash\mathbb{H}} f\, d\mu = \int_{\Gamma\backslash\mathbb{H}} \sum_{\gamma\in\langle T_1\rangle\backslash\Gamma} f(\gamma z)\, d\mu(z)$. With $f = w_t^2 b/s_\Gamma$ the inner sum is $b(z)$. □

Let M be the projection in the terminal corner onto the data g for which $\psi = w_t^{-1}Ug$ is Γ-invariant and satisfies a given system $\mathsf{D}\psi = 0$ of finite-row equations on Γ-classes. Then $UMU^* \le \iota_\Gamma\Pi_{\mathsf{D}}\,\iota_\Gamma^*$, where $\Pi_{\mathsf{D}} \in \mathcal{M}_\Gamma$ projects onto $\{\phi : \mathsf{D}(s_\Gamma^{-1/2}\phi) = 0\}$. As $U \in \mathcal{M}$ and $U^*U = I_{\mathcal{G}}$, the trace identity of Section 4 gives

$$\tau(M) = \tau(UMU^*) \le \tau_\Gamma(\Pi_{\mathsf{D}}). \tag{6.2}$$

The function $s_\Gamma^{-1/2}$ is positive, finite and constant on Γ-classes: a diagonal multiplier affiliated with $\mathcal{M}_\Gamma$ (unbounded when $N > 1$). So by the first paragraph of this subsection $\tau_\Gamma(\Pi_{\mathsf{D}})$ equals the trace of the projection onto the unweighted kernel $\{\phi : \mathsf{D}\phi = 0\}$. This is an equality of kernel dimensions, not a uniform equivalence of global Hilbert norms. It applies to the old-trace and sign equations of Section 6.1.

6.3. **Local dimensions and reversal.** Put $a_\ell = v_\ell(N)$ for a prime ℓ. Let

$$I_p(0) = 1, \qquad I_p(a) = (p+1)p^{a-1} \quad (a \ge 1),$$

the number of non-backtracking paths of length a from a fixed vertex of the p-adic tree, so that $[\mathrm{PSL}_2(\mathbb{Z}) : \Gamma] = \prod_{p|N} I_p(a_p)$ and $\mu(\Gamma\backslash\mathbb{H}) = \frac{\pi}{3}\prod_{p|N} I_p(a_p)$. Define

$$\nu_p(1) = p - 1, \qquad \nu_p(a) = I_p(a) - 2I_p(a-1) + I_p(a-2) \quad (a \ge 2), \tag{6.3}$$

$$\rho_p(a) = \begin{cases} -1, & a = 2, \\ 0, & a = 1 \text{ or } a \ge 3. \end{cases} \tag{6.4}$$

In particular $\nu_2(2) = 1$ with $\rho_2(2) = -1$, whereas $\nu_2(3) = 3$ with $\rho_2(3) = 0$. We show that $\nu_p(a_p)/I_p(a_p)$ and $\rho_p(a_p)/I_p(a_p)$ are the normalized traces of the new projection at p and of the Atkin–Lehner reversal on it, and that these traces multiply over the primes dividing N. This is a statement in $\mathcal{M}_\Gamma$, not a tensor decomposition of a classical newspace.

Let $K_0(\ell^a) \subset \mathrm{PGL}_2(\mathbb{Q}_\ell)$ be the image of the matrices in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ with lower left entry in $\ell^a\mathbb{Z}_\ell$. For a prime p and a subgroup $L \subset \mathrm{PGL}_2(\mathbb{Q}_p)$ write

$$\Gamma^{(p)} = \{g \in G_{\mathbb{Q}} : g \in K_0(\ell^{a_\ell})\ (\ell \ne p)\}, \qquad \Gamma[L] = \{g \in \Gamma^{(p)} : g \in L\}.$$

A primitive integral representative of an element of $G_{\mathbb{Q}}$ lying in every $\mathrm{PGL}_2(\mathbb{Z}_\ell)$ has determinant one. Hence $\Gamma = \Gamma[K_0(p^{a_p})]$, and for $r \ge 2$ distinct primes $p_1, \dots, p_r$

$$\Gamma^{(p_1)} \cap \langle \Gamma^{(p_2)}, \dots, \Gamma^{(p_r)}\rangle = \Gamma, \tag{6.5}$$

because every generator on the right lies in $K_0(p_1^{a_{p_1}})$.

Fix $p \mid N$, $a = a_p$, $L = K_0(p^{a-1})$, and put

$$\Delta_p^0 = \Gamma_0(N/p) = \Gamma[L], \qquad \Delta_p^1 = D_p^{-1}\Gamma_0(N/p)D_p = \Gamma[D_p^{-1}LD_p].$$

Both contain Γ with index $i_p = I_p(a)/I_p(a-1)$. Let E_p^e $(e = 0, 1)$ be the projection onto the Δ_p^e-invariant vectors, that is, the average over $\Gamma\backslash\Delta_p^e$. Their ranges are the two old ranges, and the old-trace equations of Section 6.1 read $E_p^0\phi = E_p^1\phi = 0$. Let $W_p = \begin{pmatrix} p^aA & B \\ NC & p^aD \end{pmatrix}$ have determinant p^a, and $\omega = \begin{pmatrix} 0 & -1 \\ p^a & 0 \end{pmatrix}$. Then $W_p\omega^{-1} \in K_0(p^a)$, ω normalizes $K_0(p^a)$, $\omega L\omega^{-1} = D_p^{-1}LD_p \supset K_0(p^a)$, and $W_p \in K_0(\ell^{a_\ell})$ for $\ell \neq p$. Consequently W_p normalizes Γ, $W_p^2 \in \Gamma$, $W_p\Delta_p^0W_p^{-1} = \Delta_p^1$, and $W_N \in \Gamma\prod_p W_p$.

For finitely supported c on $\Gamma\backslash\Gamma^{(p)}$ with $c(\Gamma g\gamma) = c(\Gamma g)$ for $\gamma \in \Gamma$, put

$$(S_c\phi)(\Gamma z) = \sum_{\Gamma g} c(\Gamma g)\,\phi(\Gamma gz).$$

These operators form an algebra $\mathcal{R}_p \subset \mathcal{M}_\Gamma$ containing E_p^e and W_p. On a free orbit $\Gamma g_r \cdots g_1 z = \Gamma z$ forces $g_r \cdots g_1 \in \Gamma$. If $g_j \in \Gamma^{(p_j)}$ with distinct p_j, then (6.5) gives $g_1 \in \Gamma$, and inductively every $g_j \in \Gamma$. Thus the rooted diagonal of $S_{c_1} \cdots S_{c_r}$ is $\prod_j c_j(\Gamma)$, and

$$\tau_\Gamma(X_1 \cdots X_r) = \mu(\Gamma\backslash\mathbb{H})^{1-r}\prod_j \tau_\Gamma(X_j), \qquad X_j \in \mathcal{C}_{p_j},\ p_j \text{ distinct}. \tag{6.6}$$

Let $\mathcal{A}_p$ be the von Neumann algebra generated by E_p^0, E_p^1, W_p. Polynomials in these generators lie in $\mathcal{R}_p$ and are strongly dense in $\mathcal{A}_p$. By Kaplansky density, normality and the trace property, (6.6) holds for $X_j \in \mathcal{A}_{p_j}$.

The algebras $\mathcal{A}_p$ and $\mathcal{A}_{p'}$ commute for $p \neq p'$. Indeed $\Delta = \Delta_p^e$ and $\tilde{\Delta} = \Delta_{p'}^{e'}$ lie in $\Lambda = D^{-1}\Gamma_0(N/pp')D$, $D = D_p^eD_{p'}^{e'}$, and $[\Lambda : \Gamma] = i_pi_{p'}$ by comparing covolumes. By (6.5), $\Delta \cap \tilde{\Delta} = \Gamma$, so the cosets $\Gamma\tilde{\delta}\delta$ are $i_pi_{p'}$ in number and exhaust $\Gamma\backslash\Lambda$. Hence $E_p^eE_{p'}^{e'}$ is the average over $\Gamma\backslash\Lambda$, and so is the product in the other order. Next, W_p lies in $K_0(p'^{a_{p'}})$, which is contained in both local groups defining $\Delta_{p'}^{e'}$, and it normalizes $K_0(p^a)$. So W_p normalizes $\Delta_{p'}^{e'}$ and commutes with $E_{p'}^{e'}$. The commutator of W_p and $W_{p'}$ lies in every $K_0(\ell^{a_\ell})$, hence in Γ.

Each E_p^e has rooted diagonal $1/i_p$. The meet $E_p^\wedge = E_p^0 \wedge E_p^1$ projects onto the vectors invariant under $\Delta = \langle\Delta_p^0, \Delta_p^1\rangle$. If $a = 1$, then $T_{1/p}\begin{pmatrix} 1 & 0 \\ N/p & 1 \end{pmatrix} \in \Delta$ has trace $2 + N/p^2 \notin \mathbb{Z}$. If some power of it lay in Γ, both of its eigenvalues, and hence its trace, would be algebraic integers; so no power lies in Γ and $[\Delta : \Gamma] = \infty$. On a free orbit an invariant vector is constant on the infinitely many classes $\Gamma\delta z$, and vanishes. If $a \geq 2$, put $L = N/p^2$. After conjugation by D_p, the two overgroups become

$$A = \{g \in \Gamma_0(L) : b(g) \equiv 0 \pmod p\}, \qquad B = \Gamma_0(pL).$$

If $p \nmid L$, reduction modulo p is onto $\mathrm{SL}_2(\mathbf{F}_p)$: T_1 and $\begin{pmatrix} 1 & 0 \\ L & 1 \end{pmatrix}$ give the two elementary unipotent groups. The images of A, B are opposite Borels, both contain the reduction kernel, and their images generate $\mathrm{SL}_2(\mathbf{F}_p)$. Thus they generate $\Gamma_0(L)$. If $p \mid L$, B has prime index p in $\Gamma_0(L)$, while $\begin{pmatrix} 1 & 0 \\ L & 1 \end{pmatrix} \in A \setminus B$, giving the same conclusion. Undoing the conjugation yields $\Delta = D_p^{-1}\Gamma_0(N/p^2)D_p$. Its index over Γ is $p(p+1)$ for $a = 2$ and p^2 for $a \geq 3$. Consequently $E_p^\wedge$ has rooted diagonal

$I_p(a-2)/I_p(a)$. The identity $\tau_\Gamma(E\vee E')+\tau_\Gamma(E\wedge E')=\tau_\Gamma(E)+\tau_\Gamma(E')$ for projections in a finite von Neumann algebra (closures of sums are understood; no closed-range assumption is needed) gives

$$\tau_\Gamma(P_p)=\mu(\Gamma\backslash\mathbb{H})\,\frac{\nu_p(a)}{I_p(a)},\qquad P_p=I-E_p^0\vee E_p^1,$$

with $\nu_p(a)$ as in (6.3).

For the reversal consider, on two copies of the space,

$$\mathsf{S}=\begin{pmatrix}E_p^0 & E_p^1\end{pmatrix},\quad \mathsf{F}=\begin{pmatrix}E_p^0 & 0\\ 0 & E_p^1\end{pmatrix},\quad \widehat{W}=\begin{pmatrix}0 & W_p\\ W_p & 0\end{pmatrix},\quad \mathsf{Q}=\frac{1}{2}\begin{pmatrix}E_p^\wedge & -E_p^\wedge\\ -E_p^\wedge & E_p^\wedge\end{pmatrix}.$$

Here Q is the projection onto $\ker\mathsf{S}\cap\operatorname{Ran}\mathsf{F}$. Since $W_pE_p^0=E_p^1W_p$, the operator $\widehat{W}$ commutes with F and $\mathsf{S}\widehat{W}=W_p\mathsf{S}$. The polar isometry of S therefore intertwines $\widehat{W}(\mathsf{F}-\mathsf{Q})$ with $W_p(E_p^0\vee E_p^1)$, and

$$\tau_\Gamma\big(W_p(E_p^0\vee E_p^1)\big)=(\tau_\Gamma\otimes\operatorname{tr})\big(\widehat{W}(\mathsf{F}-\mathsf{Q})\big)=\tau_\Gamma(W_pE_p^\wedge).$$

As $W_p\notin\Gamma$, $\tau_\Gamma(W_p)=0$, so $\tau_\Gamma(W_pP_p)=-\tau_\Gamma(W_pE_p^\wedge)$. This vanishes for $a=1$. For $a\geq 2$ it is $-\mu(\Gamma\backslash\mathbb{H})/[\Delta_p'':\Gamma]$ if $W_p\in\Delta_p''$ and 0 otherwise, and $\omega\in D_p^{-1}K_0(p^{a-2})D_p$ exactly when $a=2$. Hence

$$\tau_\Gamma(W_pP_p)=\mu(\Gamma\backslash\mathbb{H})\,\frac{\rho_p(a)}{I_p(a)},$$

with $\rho_p(a)$ as in (6.4).

Finally let $\sigma_p=\pm1$ and $X_p=\frac{1}{2}P_p(I+\sigma_pW_p)\in\mathcal{A}_p$, a projection because W_p commutes with P_p and $W_p^2\in\Gamma$. The X_p commute, W_N acts as $\prod_p W_p$, and by (6.6)

$$\tau_\Gamma\Big(\prod_{p\mid N}X_p\Big)=\frac{\pi}{3}\prod_{p\mid N}\frac{\nu_p(a_p)+\sigma_p\rho_p(a_p)}{2}.\tag{6.7}$$

On the range the Fricke sign is $\prod_p\sigma_p$.

6.4. **The total leading coefficient.** Set

$$\lambda(p^a)=\begin{cases}(p-1)/(p+1), & a=1,\\ (p^2-p-1)/(p(p+1)), & a=2,\\ (1-p^{-1})^2, & a\geq 3,\end{cases}\qquad c_N=\frac{I(N)}{12}\prod_{p^a\parallel N}\lambda(p^a),\tag{6.8}$$

where $I(N)=N\prod_{p\mid N}(1+p^{-1})$. These are precisely $\nu_p(a)/I_p(a)$. Let P_e be the quotient projection imposing both old traces, the signs $-\delta_p$ for $p\parallel N$, and global Fricke sign w. Write $t_e=\tau_\Gamma(\mathsf{P}_e)$. The different sign projections are orthogonal, and their sum is the common new projection. The prime-product formula therefore gives

$$0\leq t_e,\qquad \sum_e t_e=4\pi c_N.\tag{6.9}$$

The trace of the corresponding terminal incidence projection is at most t_e, by (6.2). Only this inequality is needed; no equality with a physical sector dimension is asserted.

7. Quantitative detection of the quotient target

Let $U=R(R^*R)^{-1/2}$ be the regular reconstruction isometry, let $d=\tau(I_{\mathcal{G}})$, and write $u=Ug$, $\psi=w_t^{-1}u$. Fix a finite generating set for $\Gamma=\Gamma_0(N)$. Let P be the unweighted quotient-module projection imposing the two old-trace equations at every bad prime, the prescribed signs at $p\parallel N$, and the prescribed global Fricke sign. Its trace is the target trace computed in the manuscript.

There are constants C, A, H_0, depending only on the fixed level, sector, certificate and generators, such that for every $H \geq H_0$ one can form a finite compact target-residual family D_H on terminal data for which

$$\tau\big(\mathbf{1}_{[0,C^{-1}H^{-A}]}(\mathsf{D}_H^*\mathsf{D}_H)\big) \leq \tau_\Gamma(\mathsf{P}) + C/H. \tag{7.1}$$

The target residuals are fixed-generator modular defects together with the initial Fricke, bad-prime and ramified functional equations, composed with U. Before composition with U, their source and image heights lie in $[C^{-1}H^{-A}, CH^A]$, and their norms and the number of branch pieces are polynomially bounded in H. The composed terminal operators have the same polynomial norm bounds. Only the initial residuals, not the individual modular defects, are known equations for the holomorphic problem.

1. Raw sampling on a fundamental domain. Choose a finite-sided fundamental domain $\mathcal{F}_\Gamma$ and standard cusp coordinates, and let $\mathcal{F}_\Gamma(H)$ remove all cusp tails of height exceeding H. The cusp area formula gives $\mu(\mathcal{F}_\Gamma \setminus \mathcal{F}_\Gamma(H)) \ll_N H^{-1}$. All points of $\mathcal{F}_\Gamma(H)$, folded modulo T_1, have height at least c_N/H and at most $C_N H$. Define a bounded map to the unweighted quotient module by

$$T_H g(\Gamma z) = \mathbf{1}_{\mathcal{F}_\Gamma(H)}(z)\psi(z), \qquad z \in \mathcal{F}_\Gamma.$$

Its norm is $O(H^t)$, since it samples distinct translation coordinates and $w_t^{-1} \ll H^t$ on this region.

Reduction of the terminal strip to $\mathcal{F}_\Gamma$ uses finitely many fixed elements of Γ. Indeed, the sufficiently high infinity cusp is already in this fundamental domain, and the remaining terminal strip is compact, where the modular tessellation is locally finite. Fix words in the generators for those finitely many elements. The terminal restriction of Ug is $(R^*R)^{-1/2}g$, of norm at least $\|R\|^{-1}\|g\|$. Word telescoping on the compact reduction pieces therefore gives

$$\|g\|^2 \leq C\left(\|T_H g\|^2 + \|\mathsf{D}_H g\|^2 + \|E_{\infty,H} g\|^2\right). \tag{7.2}$$

Here $E_{\infty,H} g = \mathbf{1}_{y>cH} Ug$ is restriction of the reconstructed vector to the high terminal strip, for a fixed $c > 0$, and its target module has trace $O(H^{-1})$. The constants in this paragraph can be independent of H; enlarge H_0 so all compact reduction pieces are retained.

2. The fixed quotient spectral gap. On the unweighted quotient take the finite row operator

$$\mathsf{L} = \big((E_p^0, E_p^1)_{p|N}, (W_p - s_p)_{p\|N}, W_N - w\big).$$

Then $\ker \mathsf{L} = \operatorname{Ran} \mathsf{P}$ and

$$\mathsf{L}^*\mathsf{L} \geq \gamma_N (I - \mathsf{P}) \tag{7.3}$$

for a fixed $\gamma_N > 0$. For exponent one, the two finite-index extensions of Γ are the two vertex stabilizers adjoining an edge in the $(p+1)$-regular Bruhat–Tits tree. The edge stabilizer is Γ; reduction modulo p of $\Gamma_0(N/p)$ is transitive on the $p+1$ neighbors, so the generated group acts on the full tree. To check the stabilizers precisely, put $\Delta = \langle \Delta_p^0, \Delta_p^1 \rangle$. All its elements satisfy the fixed local $K_0(\ell^{a_\ell})$ conditions at $\ell \neq p$. An element fixing the first vertex also lies in $\mathrm{PGL}_2(\mathbf{Z}_p)$; the primitive-integral-representative argument then identifies its stabilizer with $\Gamma_0(N/p) = \Delta_p^0$. Conjugating by D_p gives the second stabilizer Δ_p^1. Fixing both vertices imposes $K_0(p)$ at p, so the edge stabilizer is exactly $\Gamma_0(N) = \Gamma$. The action preserves the two vertex types because the generators have determinant of even p-adic valuation, so no edge inversion occurs. Finally the reduction of Δ_p^0 modulo p contains both elementary unipotent subgroups: use T_1 and $\left(\begin{smallmatrix} 1 & 0 \\ N/p & 1 \end{smallmatrix}\right)$, since $p \nmid N/p$. They generate $\mathrm{SL}_2(\mathbf{F}_p)$, which is transitive on $\mathbf{P}^1(\mathbf{F}_p)$. The conjugate statement holds at the other vertex. Inducting along paths therefore shows that Δ is transitive on all edges and on each vertex type of the tree. Thus each Δ-orbit of quotient coordinates is exactly the edge-coset model used in the averaging argument, with no extra stabilizer or quotient graph. On each orbit

the angle between the two vertex-averaging ranges has cosine at most $2\sqrt{p}/(p+1)$, the normalized adjacency norm of this regular tree. Thus

$$E_p^0 + E_p^1 \geq \left(1 - \frac{2\sqrt{p}}{p+1}\right)(I - P_p).$$

For completeness, this upper norm bound is elementary. Orient the tree toward one fixed end; each vertex has one parent and p children. Cauchy–Schwarz gives $|\sum_v \sum_{u \text{ child of } v} \overline{f(v)} f(u)| \leq \sqrt{p}\,\|f\|_2^2$, so the adjacency norm is at most $2\sqrt{p}$. The normalized lifts from either vertex type to edges are isometries J_0, J_1 with $J_0^* J_1$ equal to the bipartite adjacency block divided by $p+1$. The Gram matrix of $(J_0\ J_1)$ is $\left(\begin{smallmatrix} I & J_0^* J_1 \\ J_1^* J_0 & I \end{smallmatrix}\right)$, bounded below by $1 - 2\sqrt{p}/(p+1)$. Its nonzero spectrum is that of $E_p^0 + E_p^1$, proving the displayed gap. For every higher exponent $a \geq 2$, the generated group has finite index over Γ, namely $p(p+1)$ for $a = 2$ and p^2 for $a \geq 3$, as proved in Section 6.3. The same claim follows from the positive minimum of the nonzero spectrum of the corresponding fixed finite averaging matrix. The local sums $E_p^0 + E_p^1$ commute with each other and all the sign terms. The sign terms have nonzero eigenvalue 4, so their sum has a fixed gap off the common kernel. This proves (7.3).

3. Finite local equations and boundary rows. There is a finite compact target family D_H, a bounded operator A_H of norm $O(H^A)$, and a bounded boundary map B_H such that

$$\mathsf{L} T_H = A_H \mathsf{D}_H + B_H, \qquad \tau(\text{target module of } B_H) \ll_N H^{-1}. \tag{7.4}$$

Here is how to construct them, avoiding an uncontrolled infinite word argument. Every row of L is a fixed finite sum of rational branches. For a fixed rational branch, reduction of its image back to $\mathcal{F}_\Gamma$ has finitely many pieces modulo Γ: away from the finitely many rational cusps this follows from local finiteness, and near a cusp the map in cusp coordinates is upper triangular and changes height by a fixed positive factor. Consequently a uniform constant C_0 bounds the cusp-height change for every branch of L. More explicitly, choose cusp scaling matrices σ_a, σ_b and $\gamma \in \Gamma$ so that $\gamma g a = b$. Then $\sigma_b^{-1} \gamma g \sigma_a$ is upper triangular and acts in cusp coordinates by $z \mapsto \alpha z + \beta$, where $\alpha > 0$ and α, β are fixed. The source cusp has horizontal coordinate in a fixed bounded interval of its cusp width. Its image horizontal coordinate therefore also stays in a fixed bounded interval, independently of cusp height. Only finitely many target cusp translations are required to return that interval to the chosen fundamental cusp interval. This supplies the finite reduction pieces near each cusp, while $\Im z$ changes by the fixed factor α.

Retain the rows based at $\mathcal{F}_\Gamma(H/C_0)$. All their sampled quotient coordinates lie in $\mathcal{F}_\Gamma(H)$, so no input is removed by T_H. Replace their reduced quotient values by actual ψ-values using fixed words for the finitely many reducing elements of Γ. The discrepancy is a finite sum of generator defects. The old-trace and sign identities of the manuscript then express the resulting rows in terms of the initial residuals and further generator defects. These are finite group-algebra identities: in particular the sole inverse R_p^{-1} is the fixed polynomial $(R_p - (p-2)I)/(p-1)$. All sources, targets and intermediate points are in compact rational images of $\mathcal{F}_\Gamma(H)$, hence have heights bounded above and below by fixed powers of H. Removing the positive coordinate weights thus costs only H^A. Stack these compact residuals as D_H and their finite recombination as A_H.

All omitted output rows are based in quotient cusp height $> H/C_0$. Their direct sum, with one label per fixed row, has trace $O_N(H^{-1})$; define B_H to be the omitted part of $\mathsf{L} T_H$. This proves (7.4). No estimate on the norm of B_H is needed for the rank argument, only boundedness for fixed H.

4. Spectral conclusion. Combine (7.2), (7.3), and (7.4) to obtain, after increasing C, A,

$$\|g\|^2 \leq C H^A \left(\|\mathsf{P} T_H g\|^2 + \|\mathsf{D}_H g\|^2 + \|B_H g\|^2 + \|E_{\infty,H} g\|^2 \right).$$

The map $Z_H = (\mathsf{P} T_H, B_H, E_{\infty,H})$ has range trace at most $\tau_\Gamma(\mathsf{P}) + C/H$. On $\ker Z_H$ the last display gives $\|\mathsf{D}_H g\|^2 \geq (CH^A)^{-1} \|g\|^2$. The low spectral subspace for $\mathsf{D}_H^* \mathsf{D}_H$ below $(2CH^A)^{-1}$ therefore

has zero intersection with $\ker Z_H$. The finite von Neumann dimension inequality bounds its trace by the range trace of Z_H, which proves (7.1) after renaming C.

7.1. **Density-one single-prime certificates.** Write

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & 0 \\ N & 1 \end{pmatrix}, \qquad G_1 = \Gamma_1(N) \subset G_0 = \Gamma_0(N).$$

In even weight these groups may equivalently be replaced by their projective images. Fix generators $\gamma_1, \dots, \gamma_t$ of G_1. The arithmetic condition used below is that every unit class modulo a fixed M contains a positive divisor of $p - Q$. Section 8 proves that failure has summable reciprocal prime mass. Its sieve bound also shows that the primes satisfying any fixed finite list of these conditions have relative density one.

Lemma 7.1 (Polynomial certificates at a density-one set of primes)**.** *There is a set $\mathcal{Q}$ of primes, of relative density one, such that for every sufficiently large $q \in \mathcal{Q}$, each fixed γ_i-defect of a T-periodic section F is a linear combination of at most $C_N q$ translates of the Fricke defect $E = F|W_N - wF$, and of a bounded number of individual prime-q defects $D_r = wF|g_r - F$. The coefficients are bounded by a constant depending only on the fixed data. Every translating matrix has rational height at most $C_N q^{A_N}$, and determinant and inverse height bounded by fixed powers of q. In particular, the associated compact norm comparison costs only a fixed power of qH on sets of polynomial H-geometry.*

Proof. For $N = 1$, the two elementary unipotents T, V generate G_1, and $V = W_N T^{-1} W_N$ projectively. The conclusion follows from periodicity and the Fricke residual alone. Assume $N \geq 2$.

For each unit a mod N, choose a fixed $\delta_a \in G_0$ whose upper-left entry is congruent to a. Such a matrix exists by Bezout. Form the fixed list

$$h_{a,0} = \delta_a, \qquad h_{a,i} = \delta_a \gamma_i \quad (1 \leq i \leq t).$$

Every upper-left entry $A_{a,i}$ is congruent to a modulo N, and is nonzero. If the lower-left entry is $NC_{a,i}$, then $\gcd(A_{a,i}, C_{a,i}) = 1$. Left multiplication by V^n replaces $C_{a,i}$ by $C_{a,i} + nA_{a,i}$. By Dirichlet's theorem, choose fixed integers $n_{a,i}$ making these numbers distinct positive primes $c_{a,i}$ not dividing N. Their coprimality with $A_{a,i}$ persists. The Chinese remainder theorem now gives a fixed positive integer Q_a with

$$Q_a \equiv a \pmod{N}, \qquad Q_a \equiv A_{a,i} \pmod{c_{a,i}} \quad (0 \leq i \leq t).$$

Thus $\gcd(c_{a,i}, NQ_a) = 1$. Put

$$t_{a,i} = \frac{Q_a - A_{a,i}}{Nc_{a,i}}, \qquad \widehat{h}_{a,i} = T^{t_{a,i}} V^{n_{a,i}} h_{a,i} = \begin{pmatrix} Q_a & B_{a,i} \\ Nc_{a,i} & D_{a,i} \end{pmatrix}.$$

All quantities in this display are fixed, independently of q.

For each a apply the divisor-density assertion with $(Q, M) = (Q_a, NQ_a)$, and intersect its finitely many density-one sets. Remove primes dividing the fixed reconstruction certificate, N, or any fixed initial modulus, as well as finitely many small primes. The resulting set $\mathcal{Q}$ still has relative density one. If $q \in \mathcal{Q}$, choose $a \equiv q \pmod{N}$. For every i choose a positive divisor $d_{q,i}$ of $q - Q_a$ such that

$$d_{q,i} \equiv c_{a,i} \pmod{NQ_a}.$$

We arranged $q > Q_a$, so $1 \leq d_{q,i} < q$. Since $\gcd(d_{q,i}, N) = 1$ and $N \mid q - Q_a$, define integers

$$r_{q,i} = \frac{d_{q,i} - c_{a,i}}{Q_a}, \qquad s_{q,i} = \frac{q - Q_a}{Nd_{q,i}}.$$

They satisfy $|r_{q,i}| + |s_{q,i}| \ll_N q$. Multiplication gives

$$M_{q,i} := T^{s_{q,i}} V^{r_{q,i}} \widehat{h}_{a,i} = \begin{pmatrix} q & B'_{q,i} \\ Nd_{q,i} & D'_{q,i} \end{pmatrix} \in G_0.$$

All its entries have size $O_N(q^2)$; this deliberately loose bound suffices.

Write $\alpha = q - d_{q,i} \in \{1, \dots, q-1\}$, choose $\beta \in \{1, \dots, q-1\}$ with $\beta \equiv -B'_{q,i} \pmod q$, and put $l = (-\beta - B'_{q,i})/q = O_N(q)$. The determinant identity shows that $B'_{q,i}$ is nonzero modulo q. Then

$$C_{q,i} := V^{-1} M_{q,i} T^l = \begin{pmatrix} q & -\beta \\ -N\alpha & (1 + N\alpha\beta)/q \end{pmatrix}, \qquad N\alpha\beta \equiv -1 \pmod q.$$

This is one of the canonical prime-q matrices, and $C_{q,i} = W_N g_\alpha^{-1}$ as in (5.1). Consequently

$$F|C_{q,i} - F = (E - wD_\alpha)|g_\alpha^{-1}.$$

Here are the defect identities needed to recover the fixed generators. Set $\Delta_h = F|h - F$. For any integer j, Δ_{V^j} is a sum of $|j|$ translates, with signs, of

$$\Delta_V = E|T^{-1}W_N + wE.$$

For negative j this follows from $\Delta_{V^{-1}} = -\Delta_V|V^{-1}$. Exact T-periodicity gives

$$\begin{aligned} \Delta_{\widehat{h}_{a,i}} &= \Delta_{M_{q,i}} - \Delta_{V^{r_{q,i}}}|\widehat{h}_{a,i}, \\ \Delta_{M_{q,i}} &= (\Delta_V|C_{q,i} + \Delta_{C_{q,i}})|T^{-l}, \\ \Delta_{h_{a,i}} &= \Delta_{\widehat{h}_{a,i}} - \Delta_{V^{n_{a,i}}}|h_{a,i}, \\ \Delta_{\gamma_i} &= \Delta_{h_{a,i}} - \Delta_{h_{a,0}}|\gamma_i. \end{aligned}$$

Every displayed identity holds before imposing any modular or Fricke equation. The first and third identities use only that the leftmost powers of T disappear when applied to F.

The resulting expression for Δ_{γ_i} uses at most two individual D_r terms and $O_N(q)$ translates of E. Powers of V occurring in telescoping have exponent $O_N(q)$; every other factor has height $O_N(q^2)$, and each translating matrix is a product of a bounded number of these factors. Thus all matrices, including intermediate ones, have polynomial height. It is essential here that the $O(q)$ repeated factors are contiguous powers of one unipotent: their partial products have polynomial, rather than exponential, height. The standard rational-image bounds on real and imaginary parts, followed by Cauchy–Schwarz over $O_N(q)$ summands, prove the compact norm assertion. □

Fixed prime equations remove the diamond quotient. Let $F|T_1 = F$, $E = F|W_N - wF$, and fix a prime $\ell \nmid N$. Write the individual defects and their known averaged equation as

$$D_a = wF|g_a - F, \qquad B_\ell(F) = \sum_{a=1}^{\ell-1} D_a|T_{a/\ell}.$$

Let $C_a = W_N g_a^{-1} \in \Gamma_0(N)$ be the canonical matrices; all have upper-left entry ℓ. Set $g = g_1$ and $D = D_1$. Because $\Gamma_1(N)$ is the kernel of the upper-left residue homomorphism on $\Gamma_0(N)$,

$$\kappa_a = g_a g^{-1} = C_a^{-1} C_1 \in \Gamma_1(N).$$

Moreover W_N normalizes $\Gamma_1(N)$, as is seen directly from

$$W_N \begin{pmatrix} a & b \\ Nc & d \end{pmatrix} W_N^{-1} = \begin{pmatrix} d & -c \\ -Nb & a \end{pmatrix}.$$

Consequently $g = C_1^{-1} W_N$ normalizes $\Gamma_1(N)$, and $\kappa = g T_1 g^{-1} \in \Gamma_1(N)$.

For $\Delta_h = F|h - F$, put

$$x = T_{1/\ell}, \qquad K = \sum_{a=1}^{\ell-1} x^a, \qquad P = \frac{K - (\ell-2)I}{\ell - 1}, \qquad G = w \sum_{a=1}^{\ell-1} \Delta_{\kappa_a}|gx^a.$$

Here x, K, P are elements of the rational group algebra of translations, acting on the right. Since $D_a - D = w\Delta_{\kappa_a}|g$,

$$B_\ell(F) = D|K + G.$$

The elementary polynomial identity modulo $x^\ell - 1$ is $K^2 - (\ell-2)K - (\ell-1)I = 0$. Therefore a fixed polynomial $Q_\ell(x)$ satisfies

$$KP - I = (x^\ell - I)Q_\ell(x) = (T_1 - I)Q_\ell(x).$$

Periodicity of F now gives

$$D|(T_1 - I) = w\Delta_\kappa|g,$$

and hence the following identity holds without assuming any modularity or Fricke equation:

$$D = B_\ell(F)|P - G|P - w\Delta_\kappa|gQ_\ell(x), \qquad \Delta_{C_1} = (E - wD)|g^{-1}. \tag{7.5}$$

In particular, the C_1-defect is a fixed finite combination of translates of the original averaged prime-ℓ residual, the Fricke residual, and fixed $\Gamma_1(N)$-defects. Each of the latter is a finite combination of translates of a chosen fixed generator family, by word telescoping.

Choose finitely many primes $\ell_j \nmid N$ whose residue classes generate $(\mathbf{Z}/N\mathbf{Z})^\times$, using Dirichlet's theorem. Discard these finitely many primes from $\mathcal{Q}$. Their canonical matrices $C_{\ell_j,1}$ generate the quotient $\Gamma_0(N)/\Gamma_1(N)$. Thus a fixed generating set of $\Gamma_0(N)$ is expressible in fixed words in the $C_{\ell_j,1}$ and the chosen $\Gamma_1(N)$ generators. Equation (7.5) and word telescoping express every fixed $\Gamma_0(N)$ generator defect in the latter defects and the fixed original residuals $E, B_{\ell_1}, \dots, B_{\ell_s}$.

A common initial observation rectangle. Let D_H be the original target-residual family: fixed G_0-generator defects and fixed initial equations on polynomially many compact pieces of polynomial H-geometry before composition with U. The finite diamond factorization above and word telescoping first express its modular defects in fixed G_1-generator defects and the original residuals $E, B_{\ell_1}, \dots, B_{\ell_s}$. Apply Lemma 7.1 to the resulting fixed G_1 defects. For $q \in \mathcal{Q}$, $q \le X$, let X_q be the individual prime defect column on a common rectangle of polynomial (XH)-geometry. Choose $J_{0,X,H}$ to contain the original initial equations, including the Fricke equation and the finitely many original prime equations $B_{\ell_1}, \dots, B_{\ell_s}$, on a possibly larger rectangle of the same type. Then the lemma gives

$$\|\mathsf{D}_H g\|^2 \le C_N(XH)^{A_N}\big(\|X_q g\|^2 + \|J_{0,X,H} g\|^2\big) \qquad (q \in \mathcal{Q},\ q \le X). \tag{7.6}$$

All matrices and polynomials in the diamond factorization are fixed, so this extra step costs only a fixed power of H. In particular the existing $\Gamma_0(N)$ compact target observability applies unchanged; no new quotient spectral-gap assertion is needed.

The fixed initial stack contains only original residuals restricted to this enlarged rectangle. The q-dependent translating matrices from the proof are used for norm comparison; they are not inserted as branches of $J_{0,X,H}$. Hence the initial stack and its functional calculus remain integral at all fresh primes. The reconstruction isometry is the fixed one; its certificate may be enlarged by the finite set of initial prime moduli.

For the remaining argument fix a sector e and write $k = 2(h+1)$, $d_e(h) = \dim V_{2(h+1)}(e)$. Use $\epsilon = \varepsilon$ for the terminal height and let $t_e = \tau_\Gamma(\mathsf{P}_e)$ be the quotient trace defined above. Target observability gives its spectral excess a tail b_e/H. All constants below are uniform in h.

8. Summability of the non-detecting primes

The arithmetic preparation asks for a finite list of shifted-divisor conditions. For fixed positive integers Q, M and a unit class $a \pmod M$, call a prime $p > Q$ exceptional if there is no positive divisor $d \mid p - Q$ with $d \equiv a \pmod M$.

Lemma 8.1 (Shifted-divisor exceptions)**.** *For every such Q, M, a, the exceptional primes satisfy*

$$\sum_{p \ mexceptional} \frac{1}{p} < \infty. \tag{8.1}$$

The same conclusion holds for failure of any one condition in a fixed finite list, and after adding finitely many excluded primes.

Proof. The upper-bound sieve gives the following estimate [5]: there are $C > 0$ and $c_M > 0$ such that, for integers $Y \geq 4$, $z \geq Y^{24}$,

$$E_{Q,M,a}(x) \leq \frac{Cx}{(\log Y)^{1+c_M}} + z^6 + (Y+1) + Q, \tag{8.2}$$

where $E_{Q,M,a}(x)$ counts the exceptional primes less than x. Only positivity of c_M is needed.

For clarity, the extra logarithmic power in this input comes from a two-residue sieve. Away from the finitely many primes dividing $2Q$, an exceptional prime p avoids $0 \pmod{\ell}$ for every small prime ℓ; if $\ell \equiv a \pmod{M}$, it also avoids $Q \pmod{\ell}$. The sieve upper bound is a constant times

$$x \prod_{\ell < Y,\, \ell \nmid 2Q} \left(1 - \frac{1 + \mathbf{1}_{\ell \equiv a \ (M)}}{\ell}\right)$$

plus the error displayed in (8.2). The baseline Euler product contributes $(\log Y)^{-1}$, and the proved reciprocal-prime lower bound in the unit class contributes $(\log Y)^{-c_M}$. To see the error term explicitly, use upper-bound Selberg weights λ_d, supported on $d < z$ with $\lambda_1 = 1$ and $|\lambda_d| \leq d$, for the forbidden residues modulo primes below Y. Their quadratic main term is at most a fixed multiple of the displayed product when $z \geq Y^{24}$ (the upper-bound fundamental lemma in sieve dimension at most two). For each pair d, e the counting remainder is at most the number of forbidden residues modulo $[d, e]$, which is at most $[d, e] \leq z^2$. There are at most z^2 pairs and each weight product is at most z^2, giving the deliberately loose error z^6. The prime reciprocal estimate in a fixed unit progression is $\sum_{\ell < Y,\, \ell \equiv a \ (M)} 1/\ell = \varphi(M)^{-1} \log\log Y + O_M(1)$; finite excluded primes affect only the constant. This proves the additional logarithmic saving, with any fixed $0 < c_M < 1/\varphi(M)$. The estimate includes the small primes and the finite exclusions in its error term.

Here is the full summability deduction, with compatible integer cutoffs. Let $X_j = 2^{288(j+1)}$. Apply (8.2) with

$$x = X_{j+1}, \qquad Y = 2^{j+2}, \qquad z = Y^{24}.$$

Then $z^6 = 2^{144(j+2)}$ and $X_{j+1}/X_j = 2^{288}$, so

$$\frac{E_{Q,M,a}(X_{j+1})}{X_j} \leq \frac{C'}{(j+2)^{1+c_M}} + \frac{3+Q}{2^j}. \tag{8.3}$$

Both series on the right converge. Since $1/p \leq 1/X_j$ for $X_j \leq p < X_{j+1}$, summing (8.3), and adding the finite initial interval, proves (8.1). A finite union is bounded by the sum of its individual reciprocal sums. □

The matrix construction of Lemma 7.1 supplies a single preparation $\mathcal{G}$ of detecting primes. Its shifted-divisor conditions form a fixed finite list; for each good prime, they provide all the required matrix-generator certificates with a common polynomial height bound. The primes dividing the reconstruction denominators and the level, and any fixed initial interval, are excluded. Lemma 8.1 therefore gives

$$B_{\mathrm{bad}} := \sum_{p \notin \mathcal{G}} \frac{1}{p-1} < \infty, \tag{8.4}$$

because $1/(p-1) \le 2/p$. All sums indexed by p here and below are over primes. Take $\mathcal{G}$ to be the set $\mathcal{Q}$ defined in that construction, with its finite excluded set. Thus preparation and summability concern exactly the same primes.

Lemma 8.2 (Exact endpoint product)**.** *For every real cutoff $x \ge 3$,*

$$\prod_{\substack{p<x \\ p\in\mathcal{G}}} \left(1-\frac{1}{p}\right) \ll_N \frac{1}{\log x}. \tag{8.5}$$

Proof. The reciprocal full Euler product contains the reciprocals of all positive integers below the cutoff. Consequently, at integer cutoffs,

$$\prod_{p<n+1} (1-1/p)^{-1} \ge \sum_{m=1}^{n} \frac{1}{m} \ge \log(n+1).$$

Moreover,

$$\log(1-1/p)^{-1} = \log\left(1+\frac{1}{p-1}\right) \le \frac{1}{p-1}.$$

Removing the non-detecting factors thus multiplies the full product by at most $\exp(B_{\mathrm{bad}})$. This proves the integer version of (8.5); rounding the cutoff changes only an absolute constant. The fresh-prime tolerance is chosen as $1/p$, so its contraction factor is exactly

$$1-\frac{1-1/p}{p-1} = 1-\frac{1}{p}. \tag{8.6}$$

□

Relative density one alone does not imply (8.4). The finite exceptional reciprocal budget is the arithmetic fact which allows the exact exponent one in (8.5).

9. Prime blocks with summable crossing loss

Let

$$v(m) = \prod_{\substack{p<m+2 \\ p\in\mathcal{G}}} (1-1/p), \qquad m \ge 0.$$

Then $v(0) = 1$, v is positive and nonincreasing, and

$$v(m+1) \ge v(m)\left(1-\frac{1}{m+2}\right), \qquad v(m) \le \frac{C_0}{\log(m+2)} \tag{9.1}$$

for one $C_0 > 0$. Define a_j to be the first m with $v(m) \le 4^{-j}$. The bound in (9.1) guarantees existence and gives $a_j \le \lceil \exp(C_0 4^j) \rceil$.

Lemma 9.1. *The disjoint blocks $\mathcal{B}_j = \{p \in \mathcal{G} : a_j + 2 \le p < a_{j+1} + 2\}$ satisfy*

$$\beta_j := \prod_{p\in\mathcal{B}_j} (1-1/p) \le \frac{1}{4}\exp(2\cdot 4^{-j}), \qquad \max_{p\in\mathcal{B}_j} p \le 3\exp(C_0 4^{j+1}). \tag{9.2}$$

In particular, the sum of the logarithmic crossing losses is at most $\sum_{j\ge 0} 2\cdot 4^{-j} = 8/3$.

Proof. Induction in (9.1) gives $v(m) \ge 1/(m+1)$. Thus $1/(a_j+1) \le 4^{-j}$. For $j > 0$, minimality of a_j and the one-step lower bound yield

$$v(a_j) \ge 4^{-j}(1-4^{-j}).$$

Since $\beta_j = v(a_{j+1})/v(a_j)$, for $j \geq 1$ we obtain

$$\beta_j \leq \frac{1}{4(1-4^{-j})} \leq \frac{1}{4}\exp(2\cdot 4^{-j}),$$

using $-\log(1-u) \leq 2u$ for $0 \leq u \leq 1/2$. For $j = 0$ use $v(a_0) = 1$ and $v(a_1) \leq 1/4$. The upper bound on a_{j+1} gives the prime-size assertion, allowing for the integer rounding. □

This choice of blocks avoids charging a fixed rounding loss at every stage. The non-detecting reciprocal budget has already been absorbed once into C_0; the subsequent crossing losses form a geometric series.

10. Positive filters from fresh-prime blocks

10.1. **Triangular spectral deflation.** Let $(\mathcal{A}, \tau)$ be a finite von Neumann algebra and put $T = \tau(I)$. Rectangular operators into other trace modules are allowed. For $H \geq 0$ write $n_H(a) = \tau(\mathbf{1}_{[a,\infty)}(H))$.

Lemma 10.1 (Logarithm of an invertible triangular operator)**.** *Let $R_0, \dots, R_n$ be mutually orthogonal projections and $R = \sum_i R_i$. Suppose $L \in R\mathcal{A}R$ satisfies $R_i L R_j = 0$ for $j > i$, and its diagonal blocks are positive and obey*

$$R_i L R_i \geq \sqrt{s}\, R_i \qquad (0 < s \leq 1).$$

Then L is invertible in $R\mathcal{A}R$ and

$$\tau\big(\log(L^*L)\big) = 2\sum_i \tau\big(\log(R_i L R_i)|_{R_i}\big) \geq \tau(R)\log s. \tag{10.1}$$

Logarithms here are taken in the indicated corners.

Proof. Write $L = D + N$, with D block diagonal and N strictly lower triangular. The operator $D^{-1}N$ is nilpotent. Thus $L(t) = D + tN$ is invertible for $0 \leq t \leq 1$, with inverse given by a finite geometric series. For a differentiable positive invertible path $G(t)$ in a finite tracial algebra,

$$\frac{d}{dt}\tau(\log G(t)) = \tau(G(t)^{-1}G'(t)).$$

For completeness, differentiate the resolvent integral for $\log G$, use cyclicity, and integrate $\int_0^\infty (G + uI)^{-2}\,du = G^{-1}$. Uniform invertibility on the compact parameter interval justifies differentiation and integration. With $G(t) = L(t)^*L(t)$ this gives

$$\frac{d}{dt}\tau(\log(L(t)^*L(t))) = 2\Re\tau(L(t)^{-1}N) = 0.$$

Indeed $L(t)^{-1}N$ is strictly lower triangular, and every off-diagonal block has trace zero by cyclicity. Evaluate at $t = 0, 1$ and use the lower bound on the diagonal blocks. □

Proposition 10.2 (Collective small spectrum from deflation)**.** *Let $B_0, \dots, B_n$ be bounded residual operators and $P_{-1} = I$. Choose thresholds $s_i \geq s > 0$ and define recursively*

$$R_i = \mathbf{1}_{[s_i,\infty)}(P_{i-1}B_i^*B_iP_{i-1}) \quad \text{inside } P_{i-1}, \qquad P_i = P_{i-1} - R_i.$$

Put $P = P_n$, $R = I - P$, and $S = \sum_{i=0}^n B_i^*B_i$. For $0 < s \leq 1$, $K \geq \max(e, \|S\|)$ and $0 < \lambda < 1$,

$$\tau(\mathbf{1}_{[0,\lambda]}(S)) \leq \tau(P) + \frac{T\log(K/s)}{\log(K/\lambda)}. \tag{10.2}$$

The number of deflation steps does not appear in the numerator, except through the norm bound K.

Proof. Let $B_iR_i = V_iD_i$ be the polar decomposition, where

$$D_i = (R_iB_i^*B_iR_i)^{1/2} \geq \sqrt{s}\,R_i, \qquad V_i^*V_i = R_i.$$

Since R_i is a spectral projection of $P_{i-1}B_i^*B_iP_{i-1}$, its complementary subspace P_i has orthogonal image under B_i:

$$(B_iR_i)^*B_iP_i = R_iB_i^*B_iP_i = 0.$$

The lower bound on D_i therefore gives $V_i^*B_iP_i = 0$. Define an operator on R, identifying its mutually orthogonal output corners with R itself, by

$$L = \sum_{i=0}^{n} V_i^*B_iR\colon R \longrightarrow R.$$

Its ith output is in R_i. If $j > i$ then $R_j \leq P_i$, so $R_iLR_j = V_i^*B_iR_j = 0$, while $R_iLR_i = D_i$. Lemma 10.1 applies. Also

$$L^*L = \sum_i RB_i^*V_iV_i^*B_iR \leq RSR \leq KR.$$

Writing $r = \tau(R)$ and $a = \tau(\mathbf{1}_{[0,\lambda]}(L^*L))$ in the corner R, the spectral calculus and (10.1) yield

$$r\log s \leq \tau(\log(L^*L)) \leq a\log\lambda + (r-a)\log K.$$

Thus $a \leq r\log(K/s)/\log(K/\lambda)$.

Let $F = \mathbf{1}_{[0,\lambda]}(S)$. The finite-trace intersection inequality says $\tau(F \wedge R) \geq \tau(F) - \tau(P)$. On $F \wedge R$ the quadratic form of L^*L is at most λ times the norm squared. Min–max therefore gives $\tau(F \wedge R) \leq a$. Combining these inequalities and using $r \leq T$ proves the proposition. □

10.2. **The scheduled block update.** Fix a sector e. Work in the regular algebra over the *entire* terminal strip $0 \leq x < 1$, $y \geq \epsilon$, with hyperbolic area $dx\,dy/y^2$. Its finite trace τ satisfies

$$\tau(1) = \int_0^1 \int_\epsilon^\infty \frac{dy\,dx}{y^2} = \epsilon^{-1} =: A_\epsilon. \tag{10.3}$$

The algebra and its terminal unit are those of Section 4. In particular, starting the iteration at $F_0 = 1$ includes the complete terminal strip.

Choose a fixed $H_0 \geq \max(\epsilon^{-1}, 4)$ large enough for all the finitely many sector observation data. For a horizon n, put

$$R = 4^n, \qquad H = H_0R, \qquad T_e(H) = t_e + \frac{b_e}{H} + \frac{1}{H},$$

where $b_e \geq 0$ is the sector tail constant. This target stays fixed during all n steps of this particular iteration.

Proposition 10.3 (Actual scheduled block update)**.** *For these data there are positive contractions $1 = F_0 \geq F_1 \geq \cdots \geq F_n \geq 0$ in the terminal algebra. Their exact polynomial recipes preserve the common kernel of the residual rows in either realization. If $u_j = (\tau(F_j) - T_e(H))_+$, then, for $0 \leq j < n$,*

$$u_{j+1} \leq \left(\frac{e^{2\cdot 4^{-j}}}{4} + 2\,2^{j-n}\right) u_j + (1 + A_\epsilon\log 2)\,2^{-(n+j)}. \tag{10.4}$$

The construction is polynomial in the normalized original residual rows and the common reconstruction operators. Its total logarithmic composition budget is $O_N(4^n)$.

We give the positive update, the trace estimate, and its cost schedule. The hard spectral projections used to prove the trace bound remain in the regular representation; only polynomial expressions in known residuals are transferred to the physical representation.

For a positive contraction F, form the normalized positive Gram operator G from the actual residual rows. A soft update has the form

$$F^- = F - F\big(1 - (1 - FGF)^L\big)F,$$

with normalization ensuring $0 \le FGF \le 1$. Then $0 \le F^- \le F$. If $Fv = v$ and every residual row kills v, then $FGFv = 0$, hence $F^-v = v$. A scalar polynomial z with $0 \le z(t) \le t$ on $[0,1]$, $z(0) = 0$, and $z(1) = 1$ sharpens F^- while retaining both assertions. The Bernstein sharpening lemma provides, for integer sharpness $J \ge 1$,

$$\deg z \le 16J^5 + 1, \quad \sum_i |z_i| \le (16J^5 + 2)3^{16J^5}, \quad \tau\big(z(F^-) - z(F^-)^2\big) \le 4A_\epsilon/J. \tag{10.5}$$

The spectral comparison of the normalized Gram operator with the successive fresh-prime deflations gives contraction β_j on the trace excess over $T_e(H)$. Freshness is valid because the blocks are disjoint and preceding operators remain integral at all later primes. The generator certificates and the sector observation estimate supply the actual spectral threshold, including the ramified and Fricke rows. Thus this comparison uses the actual quotient-incidence target t_e.

If $M_j \ge 0$ is the logarithmic cost of that normalized threshold, choose local effort

$$E_j = (4^j + M_j)2^{n-j}. \tag{10.6}$$

The comparison's relative threshold loss is at most $M_j/E_j \le 2^{j-n}$. Choose the attenuation power and incoming sharpness so that the attenuation costs at most $1/E_j$ and the amplified incoming defect costs at most $1/E_j$. The normalized logarithmic trace comparison then gives exactly

$$u_{j+1} \le (\beta_j + 2\,2^{j-n})u_j + (1 + A_\epsilon \log 2)2^{-(n+j)},$$

since $E_j \ge 2^{n+j}$. This is (10.4). The scalar sharpening for the outgoing step is chosen using the next block's effort, so the defect bound is available when needed.

Polynomial observation costs and Lemma 9.1 give $M_j \le C(4^j + n + 1)$ when $H = H_0 4^n$. Consequently

$$\begin{aligned} \sum_{j<n} E_j &\le (1 + C)4^n + 2C(n + 1)2^n \\ &\le (1 + 3C)4^n. \end{aligned} \tag{10.7}$$

The second inequality uses $n + 1 \le 2^n$. The degree, coefficient, and propagated residual estimates obtained from (10.5) give the stated logarithmic composition budget. The retained-corner trace estimate below proves the spectral comparison used here. Preservation applies to every original sector equation and is not an additional hypothesis on $\mathcal{E}(N, k)$.

10.3. **The retained-corner logarithmic estimate.** We prove the excess-dependent estimate needed for the scheduled update. Work in the finite terminal algebra, with $D = \tau(1)$, and let $0 \le F \le 1$ be integral at every prime of the next block. Write

$$\delta = \tau(F - F^2), \quad P = \mathbf{1}_{[1/2,1]}(F), \quad m = \tau(P).$$

Inside P, first deflate the initial residuals at a sufficiently small threshold. The target observability estimate (7.1), together with (7.6), then applies on each remaining projection: its individual prime-defect Gram operator has high spectral mass at least the dimension excess above $T = t_e + (b_e + 1)/H$.

For the prime p, apply (5.4) with $\theta = 1/p$. The known averaged residual removes at least $1/p$ of that excess. The hard survivor P' of the whole block, of trace r, consequently satisfies

$$r - T \le \beta(m - T), \qquad \beta = \prod_{p \text{ in the block}} (1 - 1/p). \tag{10.8}$$

If $m < T$, this follows already from $r \le m$; otherwise it follows by iterating the nonnegative excess estimate. Freshness persists under these spectral cuts because earlier operators are integral at all unused primes. The initial observation rectangles are common to the block and contain only the fixed original residuals.

Normalize the sum G of the squared residuals of the block so that $0 \le G \le 1$. All hard thresholds may be taken at least $c^2 > 0$, where $\log(2/c^2) \ll_N \log(XH)$ and X bounds the primes in the block. Indeed target observation, generator recombination, residual normalization, and the tolerance $1/p$ all have polynomial costs in XH. The same geometric normalization bounds the physical Gram operator by one.

Fix $0 < d < 1$ and $K \ge 1 + d^2$. Apply the triangular construction of Lemma 10.1 in the corner P, adjoining the row dP' on the final survivor. The resulting triangular operator L has diagonal bounded below by c on the removed corners and by d on P'. Moreover $L^*L \le PGP + d^2P$. Operator monotonicity of the logarithm gives

$$\tau_P \log(PGP + d^2P) \ge (m - r) \log c^2 + r \log d^2.$$

If $b = \tau_P(\mathbf{1}_{[0,d^2]}(PGP))$, splitting this logarithm at d^2 gives the upper bound $b \log(2d^2) + (m-b) \log K$. Hence, with $D_0 = \log(K/(2d^2)) > 0$ and $\Theta = \log(K/c^2)/D_0$, we have

$$b \le r + (m - r)\Theta + r\frac{\log 2}{D_0}. \tag{10.9}$$

The factor $m - r$ is essential: replacing it by the full terminal trace would lose the endpoint schedule.

Set $C = FGF$ and $a = e^{-Ld^2/32}$ for an integer $L \ge 1$. On P, F is invertible with inverse norm at most two. Pulling a high spectral subspace of PGP back by this inverse and applying min–max shows

$$\tau(\mathbf{1}_{[d^2/32,\infty)}(C)) \ge m - b.$$

For $h_L(t) = 1 - (1 - t)^L$, define $F^- = F - Fh_L(C)F$. Then $0 \le F^- \le F$, and $F^-v = v$ whenever $Fv = v$ and all the residuals kill v. Since $h_L(C) \le LC \le LF^2$, cyclicity and $t^2(1 - t^2) \le 2t(1 - t)$ imply

$$\tau(F) - \tau(F^-) \ge (1 - a)(m - b) - 2L\delta.$$

Any further sharpening $F^+ = z(F^-) \le F^-$ preserves this inequality. Combining it with (10.8), (10.9), $r \le D$ and $|m - \tau(F)| \le 2\delta$ yields

$$\begin{aligned} \tau(F^+) - T \le [1 - (1 - a)(1 - \beta)(1 - \Theta)](\tau(F) - T) \\ + (4 + 2L)\delta + D \log 2/D_0. \end{aligned} \tag{10.10}$$

We choose $0 \le \Theta \le 1$, so the coefficient is nonnegative and at most one; the same bound therefore holds for positive excesses.

Here is a polynomial sharpening with the stated quantitative cost. For an integer $J \ge 2$, partition $[1 - 1/J, 1 - 1/(2J)]$ into J intervals and choose an open interval of spectral mass at most D/J. Let b_0 be its midpoint and $w_0 = 1/(4J^2)$ its half-width. With $m_0 = 16J^5$, set

$$z(t) = t \sum_{j \ge \lceil m_0 b_0 \rceil} \binom{m_0}{j} t^j (1 - t)^{m_0 - j}.$$

The variance bound $\mathrm{Var}(\mathrm{Bin}(m_0, t)) \le m_0/4$ and Chebyshev's inequality bound each tail outside the chosen interval by $1/(4J)$. On its upper side also $1 - t \le 1/J$. Splitting the spectral integral into the

two sides and that interval therefore gives $\tau(z(F^-) - z(F^-)^2) \le 4D/J$. Endpoints of the interval obey the same tail bounds, so spectral atoms cause no exception. Moreover $0 \le z(t) \le t$, $z(0) = 0$, $z(1) = 1$, its degree is at most $16J^5 + 1$, and expanding the binomial sum bounds its coefficient mass by $(16J^5 + 2)3^{16J^5}$. For $J = 1$, use $z(t) = t$.

To obtain the schedule in Proposition 10.3, take

$$M_j = \log(2/c_j^2), \quad E_j = (4^j + M_j)2^{n-j}, \quad d_j = e^{-E_j/2}, \quad L_j = \lceil 32E_j e^{E_j} \rceil, \quad K = 2.$$

Then $D_0 = E_j$, $\Theta_j \le 2^{j-n}$ and $a_j \le 1/E_j \le 2^{-(n+j)}$. Choose the incoming sharpening $J_j > 4D(4+2L_j)E_j$; the initial $F_0 = 1$ has zero defect and needs no sharpening. Thus $(4+2L_j)\delta_j \le 1/E_j$. The coefficient in (10.10) is at most $\beta_j + a_j + \Theta_j \le \beta_j + 2\,2^{j-n}$, and its additive term is at most $(1 + D\log 2)2^{-(n+j)}$. This proves the stated recurrence. At step j the outgoing sharpening uses the next block's effort; after the last step take $J = 1$.

Finally $M_j \ll_N 4^j + n + 1$. Each update and sharpening has degree and logarithmic coefficient mass bounded by $\exp(O_N(E_j + E_{j+1} + 1))$. Under polynomial composition, degrees multiply, whereas logarithmic coefficient masses satisfy

$$\log(2 + \mathrm{mass}(p(A))) \le \deg(p)\log(2 + \mathrm{mass}(A)) + \log(2 + \mathrm{mass}(p)).$$

Together with $\sum_{j<n} E_j = O_N(4^n)$ this gives the bounds

$$\deg \le \exp(O_N(4^n)), \qquad \mathrm{mass} \le \exp(\exp(O_N(4^n)))$$

for the entire recipe. All scalar coefficients are chosen in the regular representation before the weight; the same polynomial is used physically, where its positivity and solution-preservation follow from the preceding scalar identities.

Lemma 10.4 (Finite-horizon endpoint schedule)**.** *Suppose $u_j \ge 0$, $a, b \ge 0$, and $\ell_j \ge 0$ satisfy*

$$u_{j+1} \le \left(\frac{e^{\ell_j}}{4} + a\,2^{j-n}\right) u_j + b\,2^{-(n+j)}, \quad 0 \le j < n, \qquad \sum_{j<n} \ell_j \le L_0.$$

Then

$$u_n \le \frac{e^{L_0+4a}(u_0 + 4b)}{4^n}. \tag{10.11}$$

Proof. Since $e^x \ge 1 + x$ and $e^{\ell_j} \ge 1$,

$$\frac{e^{\ell_j}}{4} + a2^{j-n} \le \frac{1}{4}\exp(\ell_j + 4a2^{j-n}).$$

Multiplying the recurrence by 4^{j+1} and iterating gives

$$4^n u_n \le \exp\left(\sum_{j<n}(\ell_j + 4a2^{j-n})\right)\left(u_0 + b\sum_{j<n} 4^{j+1}2^{-(n+j)}\right).$$

All terms are nonnegative, so replacing each partial exponential by the full exponential is legitimate. Finally,

$$\sum_{j<n} 2^{j-n} \le 1, \qquad \sum_{j<n} 4^{j+1}2^{-(n+j)} = 4\sum_{j<n} 2^{j-n} \le 4.$$

These inequalities prove (10.11). □

Corollary 10.5 (Full terminal trace bound)**.** *For every $n \ge 0$ the actual filter satisfies*

$$\tau(F_n) \le t_e + \frac{b_e + 1}{H_0 4^n} + \frac{A_*}{4^n}, \qquad A_* = e^{8/3+8}\big(A_\epsilon + 4(1 + A_\epsilon \log 2)\big). \tag{10.12}$$

Proof. Apply Lemma 10.4 to (10.4) with $a = 2$, $b = 1 + A_\epsilon \log 2$, $L_0 = 8/3$, and $u_0 \le \tau(1) = A_\epsilon$. Add the target $T_e(H)$. □

The schedule depends on the final horizon. An additive error that is merely summable in j would not suffice: after multiplication by the endpoint scale, it is $\sum 4^{j+1}\mathrm{error}_j$ that must be bounded.

11. Finite expressions and transfer to holomorphic dimension

The remaining issue is quantitative: an intrinsic regular filter must be represented by the same finite expression in both the regular and holomorphic realizations. The construction below uses the finite-branch estimate of Lemma 4.3.

Proposition 11.1 (Common finite realization)**.** *There are constants $K_a, K_c \ge 0$, fixed before n and h, with the following properties. Put $R = 4^n$ and*

$$s = \exp((K_a + 1)R). \tag{11.1}$$

Using common reconstruction accuracy $\exp(-8s)$*, the filter F_n has a finite expression T_n such that*

$$\mathrm{mass}(T_n),\ \mathrm{ht}(T_n) \le \exp(\exp(K_c R)), \tag{11.2}$$

$$0 \le T_n^{\mathrm{reg}} \le 1, \qquad \|T_n^{\mathrm{reg}} - F_n\| \le 2/R, \tag{11.3}$$

$$0 \le T_n^{\mathrm{hol}} \le 1, \qquad \left\| T_n^{\mathrm{hol}} P_e - P_e \right\| \le 2/R. \tag{11.4}$$

The regular norm bound is uniform over the base point. Here P_e is the actual normalized physical sector projection at weight $2(h+1)$. The common normalized reconstruction u is a contraction in both realizations, has regular source supported on the terminal strip, and has finite-expression mass and matrix height bounded by $\exp(\exp(K_u R))$ *for a fixed K_u.*

Here mass means the finite expression's coefficient bound, and height means a simultaneous bound on its branch matrices and inverses. The expression calculus gives, for the literal sandwich uT_nu^*,

$$\mathrm{mass}(uT_nu^*) \le \mathrm{mass}(u)^2\,\mathrm{mass}(T_n), \qquad \mathrm{ht}(uT_nu^*) \le 4\,\mathrm{ht}(T_n)\,\mathrm{ht}(u)^2. \tag{11.5}$$

Thus the sandwich also has double-exponential cost in R. Approximation errors are propagated through the complete polynomial expression, using its degree and coefficient budget. Choosing (11.1) makes their total at most $2/R$, as in (11.3) and (11.4); the physical estimate uses the exact preservation of all original sector equations.

Lemma 11.2 (Regular sandwich bound)**.** *Let $0 \le F \le 1$ in the finite terminal algebra, and let T be a positive contraction with $\|T - F\| \le \delta$ there. If u is a contraction with terminal source, then its retained regular trace obeys*

$$\mathrm{Tr}_{\mathrm{ret}}(uTu^*) \le \tau(F) + \delta A_\epsilon. \tag{11.6}$$

Proof. Positivity allows the output cutoff to be removed when taking the trace. On the terminal source, $T \le F + \delta I$, whence

$$\mathrm{Tr}(uTu^*) \le \mathrm{Tr}(uFu^*) + \delta\,\mathrm{Tr}(uu^*).$$

Cyclicity and the contraction bound give

$$\mathrm{Tr}(uFu^*) = \tau(F^{1/2}u^*uF^{1/2}) \le \tau(F), \qquad \mathrm{Tr}(uu^*) = \tau(u^*u) \le \tau(I) = A_\epsilon.$$

The finite-source trace justifies cyclicity and finiteness even when the output representation is larger. These are the positive sandwich and terminal support identities in the regular trace library. □

In particular, an operator norm error is paid against the fixed area A_ϵ, rather than against a weight-dependent dimension. The proof does not require an order comparison between noncommuting products such as TF and F.

Proposition 11.3 (Finite-weight comparison and its costs)**.** *Let M_R be a matrix-height bound for uT_nu^* and put $H_{\mathrm{out}} = \exp(16s)$. At weights satisfying the finite-branch separation condition*

$$(h+1)^{-1/3} M_R(H_{\mathrm{out}} + 2) \le 1, \tag{11.7}$$

the filter in Proposition 11.1 satisfies

$$\left(1 - \frac{2}{R}\right) \frac{d_e(h)}{2h+2} \le \frac{\tau(F_n) + 2A_\epsilon/R}{4\pi} + \mathcal{E}_{\mathrm{an}}(s,h) + \mathcal{E}_{\mathrm{tr}}(R,h). \tag{11.8}$$

There are fixed constants $C, D \ge 0$ such that

$$\mathcal{E}_{\mathrm{an}}(s,h) \le \frac{C}{R} + \frac{2\exp(\exp(DR))}{\sqrt{2h+2}}. \tag{11.9}$$

After increasing the fixed cost constants, the finite-expression trace error satisfies

$$\mathcal{E}_{\mathrm{tr}}(R,h) \le (h+1)^{-1/4} \tag{11.10}$$

whenever h is sufficiently large and $R \le c^2 \log\log(h+1)$, with $c > 0$ sufficiently small depending only on those constants. The same scale also guarantees (11.7)*.*

Here the finite-weight comparison is the actual holomorphic trace comparison, with its canonical periodization and cutoff geometry, not an assumed equality of regular and physical traces. On the physical side, (11.4) gives $\langle T_n^{\mathrm{hol}} v, v\rangle \ge (1 - 2/R)\,\|v\|^2$ on the normalized sector. Sum over an orthonormal basis and apply the reconstructed finite-weight trace theorem. On the regular side apply Lemma 11.2 with $\delta = 2/R$. This yields (11.8); the trace normalization is $1/(4\pi)$.

For completeness, the size of every error relevant to the endpoint is as follows. The retained reconstruction and mode errors have the form

$$c_1 e^{-3s} + \frac{c_2 s^4}{\sqrt{2h+2}} + \frac{\exp(c_3 s^4)}{\sqrt{2h+2}}, \tag{11.11}$$

for fixed nonnegative majorants c_i. Since $s = \exp((K_a+1)R)$, $e^{-3s} \le R^{-1}$, while the two coefficients involving s^4 are at most $\exp(\exp(DR))$ after increasing D. This proves (11.9) from the cutoff estimates below.

Use lower cutoff $Y = Y(s)$ and output height $H_{\mathrm{out}} = \exp(16s)$. The common construction gives $Y^{-1} \le \exp(C_Y s^4)$. Together with (11.5), this bounds the relevant matrix heights, periodization counts, and trace coefficients by $\exp(\exp(D'R))$ after enlarging D'. The finite-branch trace theorem has weight saving $(h+1)^{-1/3}$, subject to its separation inequality; the same estimates imply that inequality at the scale just specified. This gives (11.10). The same double-exponential bound pays both the separation condition and the finite-branch trace error.

One must retain the $\exp(Cs^4)$ form in (11.11) and in the cutoff bound. Replacing it by an extra iterated exponential would not justify the horizon used here.

11.1. **Construction of the common finite expression.** Here are the cutoff and normalization details for the preceding estimates. The horizon schedule bounds the degree and the logarithm of the expanded coefficient sum of the entire polynomial recipe by $\exp(CR)$. All primitive residuals have at most $\exp(CR)$ branch pieces, matrix height and coefficient bounds, and source and image heights and their inverses of that size. These assertions follow from the polynomial certificate bounds and the block-size estimates; composition of the scheduled recipes adds their logarithmic costs.

Increase the fixed constant in $s = \exp(C'R)$ so that the degree and logarithmic coefficient budget are at most s. For contractions, replacing each primitive by a norm-η approximation changes a polynomial expression by at most its coefficient mass times its degree times η, by telescoping each monomial. Thus accuracy e^{-8s} leaves error $O(e^{-5s})$ after the whole recipe.

Put $Y = e^{-Cs}$ for a sufficiently large fixed C, and truncate reconstruction below Y. The bound into the space with one less power of the capped weight gives $\|R - R^Y\| \ll Y$. Orthogonality of the removed outputs gives

$$G = R^*R \geq I, \qquad G_Y = R^{Y*}R^Y \geq I, \qquad \|G - G_Y\| \ll Y^2.$$

Both spectra lie in one fixed interval $[1, C_R]$. The inverse-square-root resolvent integral shows $\|G^{-1/2} - G_Y^{-1/2}\| \ll Y^2$. For $C_* > C_R$, approximate the inverse square root from below by

$$q_r(G_Y) = C_*^{-1/2} \sum_{j=0}^{r} \frac{\binom{2j}{j}}{4^j} (I - G_Y/C_*)^j.$$

The error is $O(\rho^r)$ for a fixed $\rho < 1$. Taking $r = O(s)$ makes it at most e^{-8s}, while $V = R^Y q_r(G_Y)$ remains a contraction: $V^*V = q_r(G_Y) G_Y q_r(G_Y) \leq I$. The same scalar polynomial works in both representations.

At height cutoff Y, reconstruction depth is $O(s)$ and the number of possible folds per step is $e^{O(s)}$. Fricke branches have terminal output height at most C/Y; affine ramified branches have output height at most $p\varepsilon$. Take an upper cutoff H_{out} larger than these and all observed heights, with $\log H_{\mathrm{out}} = O(s)$. Beyond it the Gram operator is the identity, the reconstruction is terminal inclusion, and every observation is zero. Keep this identity block exact, applying the polynomial normalization only to the compact block. Its regular trace is H_{out}^{-1}; its normalized holomorphic trace is $O(H_{\mathrm{out}}^{-1} + k^{-1})$ by (4.4).

The retained reconstruction has branch count, coefficient cost and matrix heights bounded by $\exp(O(s^2))$. The degree-$O(s)$ Gram normalization raises these bounds to $\exp(O(s^3))$. Substitution in the complete filter recipe, of degree at most s and logarithmic coefficient budget at most s, gives bounds $\exp(O(s^4))$ for the final compact sandwich. This is $\exp(\exp(O(R)))$, as required. If cutoffs are chosen smaller during composition, it is enough to keep $Y^{-1} \leq \exp(Cs^4)$; this remains the same cost class.

All exact and truncated positive sandwiches are bounded by a fixed multiple of RR^* or $R^Y R^{Y*}$. They therefore share the Fourier majorant (3.4). A norm error η costs at most $O(A\eta + A^{-1})$ in normalized holomorphic trace, by splitting at Ak Fourier modes; choose $A = \eta^{-1/2}$. This converts the above errors to $O(e^{-3s})$. In the regular realization the error is paid against the fixed finite terminal trace. The remaining compact operator is a finite rational-branch sum, so Lemma 4.3 bounds its normalized trace discrepancy by

$$k^{-1/2} \exp(Cs^4).$$

This proves the common finite-expression cost and the analytic bounds used above. The final rescaling by $1 + O(e^{-5s})$, if needed after truncating the polynomial recipe, makes the approximants contractions and costs another $O(e^{-5s})$. Positivity is preserved by using the same positive update and Gram recipes before expanding into branches.

12. Choosing the horizon

We now give the elementary scale calculation underlying the last comparison and then finish the proof.

Lemma 12.1 (Double-exponential costs at the endpoint scale)**.** *Let $D \geq 0$ be fixed. Choose $c > 0$ with $Dc^2 \leq 1/2$. If $L = \log\log(h+1) \geq 32$ and $R \leq c^2 L$, then*

$$\exp(\exp(DR))(h+1)^{-1/3} \leq (h+1)^{-1/4}. \tag{12.1}$$

Also $(2h+2)^{-1/2} \leq (h+1)^{-1/3}$ for $h \geq 0$.

Proof. Put $t = \log(h+1) = e^L$. Then $\exp(DR) \le \exp(L/2) = \sqrt{t}$. For $L \ge 32$ we have $\sqrt{t} \le t/12$, so the logarithm of the left side of (12.1) is at most $t/12 - t/3 = -t/4$. The second inequality follows directly from $h + 1 \ge 1$. □

There are only finitely many fixed cost constants to pay. Choose one $c > 0$ small enough for their sum; this simultaneously pays the reconstruction errors, the finite trace error, and the separation condition. Equivalently one may introduce the observation scale

$$X_c(h+1) = \exp\bigl(c\sqrt{\log\log(h+1)}\bigr), \qquad (\log X_c(h+1))^2 = c^2 L.$$

All constants are chosen before the horizon and weight.

Proposition 12.2 (Sector horizon bound)**.** *There are fixed $C_e, D_e \ge 0$ such that, whenever h is sufficiently large, $R = 4^n \le c^2 L$, and $D_e c^2 \le 1/2$,*

$$\left(1 - \frac{2}{R}\right)\frac{d_e(h)}{2h+2} \le \alpha_e + \frac{C_e}{R} + 3(h+1)^{-1/4}, \qquad \alpha_e = \frac{t_e}{4\pi}. \tag{12.2}$$

Proof. Insert Corollary 10.5 into (11.8). All terminal tail and approximation contributions are constant multiples of R^{-1}. By Lemma 12.1, the two analytic weight errors in (11.9) contribute at most $2(h+1)^{-1/4}$; (11.10) contributes one further copy. Absorb the fixed coefficients of R^{-1} into C_e. Enlarging D_e pays all scale conditions together. □

Lemma 12.3 (Preserving the leading coefficient)**.** *If $\alpha, E \ge 0$, $0 \le \delta \le 1/2$, and $(1-\delta)x \le \alpha + E$, then*

$$x \le \alpha + 2E + 2\delta\alpha.$$

Proof. Divide by $1 - \delta > 0$ and write $\alpha/(1-\delta) = \alpha + \delta\alpha/(1-\delta)$. Use $(1-\delta)^{-1} \le 2$ on the two error terms. □

Proposition 12.4 (Sector endpoint)**.** *For each e there is $B_e \ge 0$ such that eventually in h,*

$$\frac{d_e(h)}{2h+2} \le \alpha_e + \frac{B_e}{\log\log(h+1)}. \tag{12.3}$$

Proof. Fix $c > 0$ as above. For all sufficiently large h, choose the integer $n \ge 0$ so that

$$4^n \le c^2 L < 4^{n+1}, \qquad L = \log\log(h+1).$$

We can also require $c^2 L \ge 16$, hence $R = 4^n \ge 4$. Therefore

$$\frac{1}{R} \le \frac{4}{c^2 L}, \qquad 0 \le \frac{2}{R} \le \frac{1}{2}.$$

Apply Lemma 12.3 to (12.2). It gives

$$\frac{d_e(h)}{2h+2} \le \alpha_e + \frac{2C_e + 4\alpha_e}{R} + 6(h+1)^{-1/4}.$$

For large h, $(h+1)^{-1/4} \le 1/L$. Thus one can take

$$B_e = \frac{4}{c^2}(2C_e + 4\alpha_e) + 6.$$

In particular, removal of the approximate-invariance factor leaves α_e itself as the leading coefficient. □

13. Classical comparison and completion of the proof

Lemma 13.1. *For every fixed N, along positive even weights,*

$$\dim S_k^{\mathrm{new}}(N) = c_N k + O_N(1), \qquad 0 \le \dim M_k^\circ(N) - \dim S_k^{\mathrm{new}}(N) \le \#\,\mathrm{Cusps}(N).$$

Every normalized cuspidal newform gives an element of $\mathcal{E}(N,k)$, and distinct newforms give distinct elements.

Proof. The full cusp dimension is $(k-1)I(N)/12 + O_N(1)$. The usual oldform decomposition has multiplicity $d(N/M)$ for newforms of conductor $M \mid N$. Inverting it with $\mu * \mu$ gives the leading coefficient

$$\frac{1}{12}((\mu*\mu)*I)(N).$$

At p^a its ratio to $I(p^a)/12$ is $(I(p)-2)/(p+1)$ when $a=1$, and $(I(p^a) - 2I(p^{a-1}) + I(p^{a-2}))/I(p^a)$ when $a \ge 2$. These simplify to (6.8). The bounded remainder follows because there are finitely many divisors of fixed N; it also follows directly from the explicit newform dimension formula [4].

Map a modular form to its constant term at each cusp, using fixed scaling matrices. The kernel is $S_k(N)$. On $M_k^\circ(N)$ its kernel is exactly $S_k^{\mathrm{new}}(N)$: both use the same two-trace conditions, and a cusp form already vanishes at 0 and ∞. Thus this map induces an injection $M_k^\circ(N)/S_k^{\mathrm{new}}(N) \hookrightarrow \mathbb{C}^{\mathrm{Cusps}(N)}$. No classification of Eisenstein packets, and no assertion of semisimplicity of their weight-two traces, is needed.

The last assertion is the usual newform Euler product and completed functional equation, with the prescribed Steinberg or zero factor at a ramified prime. Cusp forms have entire finite-order completions. A normalized simultaneous Hecke eigenbasis has distinct coefficient sequences. These statements hold at every level. □

Proof of Theorem 1.2. Proposition 2.2 and linear independence already give finiteness at every positive even weight. Apply Proposition 12.4 with $k = 2(h+1)$ to the explicitly constructed reconstruction and sum over its finitely many sectors. Since $\log\log(h+1) \asymp \log\log(2k)$ for sufficiently large k, this has the required shifted logarithmic scale. Equation (6.9) and Lemma 13.1 yield

$$\dim S_k^{\mathrm{new}}(N) \le \#\mathcal{E}(N,k) \le \sum_e \dim V_k(e) \le c_N k + O_N\left(\frac{k}{\log\log(2k)}\right).$$

Subtract $\dim S_k^{\mathrm{new}}(N) = c_N k + O_N(1)$. Replacing that dimension by $\dim M_k^\circ(N)$ costs only a bounded term, by the cusp-constant injection. This proves the theorem. The finitely many initial positive even weights can be absorbed into the implied constant, since their solution sets are finite and $k/\log\log(2k)$ is positive for $k \ge 2$. □

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